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MAGNETICALLY INDUCED FORCES ON
EPICARDIAL LEADS
EXPERIMENTAL DEVICE DEVELOPMENT AND CLINICAL TESTING

by

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A THESIS

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Abstract

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For Dr. Nichols

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Chapter 1

Introduction

Medical diagnostics heavily rely on imaging tools such as MRI, a powerful and evolving application capable of showing high contrast sensitivity to soft tissue while not depending on ionizing radiation. However, safety concerns may arise during cardiac MRI procedures, electrocardiogram (ECG) leads are generally cataloged as MRI unsafe because of the strong magnetic field and radiofrequency (RF) pulses that may interact with these leads.

More than 300,000 individuals are receiving new cardiac implantable electronic device (CIED) implants every year in the U.S. alone, reported the Heart Rhythm Society (HRS) in 2024 [6]. Some of these patients will eventually require device explantation or revision, necessitating surgery [7]. For this reason, a well established protocol to monitor cardiac rhythm exists, it involves the use of temporary epicardial leads, however there is a recognized practice when these leads are removed (only if significant resistance is encountered) to cut the wire flush with the skin, intentionally leaving a small segment inside the patient to prevent immediate complications [8, 9, 10].

Temporary epicardial pacing leads are frequently utilized as part of postoperative cardiac rhythm control procedures, for example, leads are frequently placed during

open-heart surgery to regulate cardiac rhythm postoperatively, then are typically cut at the skin surface leaving behind a short length of wire embedded in the tissue [11]. When the surgical danger of removing these leads is too high, they are left in place to prevent further complications. These cases introduce long-term repercussions, most notably the loss of access to MRI scanners.

According to ASTM F2503, a device labeled as MRI safe is expected to be non-conductive, non-electrical, non-magnetic, and to present no known hazards in the MRI environment [12]. However, this classification does not apply to all epicardial leads, as they are rarely physically characterized after removal, nor properly labeled by vendors. By the time they are cut, these leads have fulfilled their clinical purpose, and systematic safety evaluations are seldom performed.

Although retained leads are nonfunctional and relatively short, that of about 8 to 20 cm, they are frequently classified as MRI unsafe due to concerns over RF heating, magnetically induced translational forces and torques, or inadvertent cardiac stimulation [13, 14, 15]. Effectively preventing patients with CIED or cardiac surgery history, who are most likely to need a MRI scan during their lifetime, to not be able to enter MRI room.

There is still a dearth of focused studies evaluating the precise dangers associated with retained epicardial leads, despite the frequent use of such leads and their repercussions. Existing clinical recommendations tend to be too cautious and frequently disqualify these individuals from MRI eligibility even when there is little evidence to support it. Recent studies regarding this problem have shown both no image quality degradation and, most importantly, no serious safety events such as, device malfunction, myocardial injury or arrhythmias, associated with MRI examinations in patients with CIEDs abandoned leads, including epicardial leads [16, 17]. While these studies have evaluated (to some extent) the resulting consequences of RF-induced heating, mechanical forces and conductivity, additional detail is needed.

Therefore, the primary safety concerns regarding retained cardiac leads during MRI arise from three main interaction mechanisms. First, the strong magnetic field and its spatial gradient can exert translational forces and torques on the lead, resulting in mechanical displacement along the direction of the magnetic field. Second, the lead can absorb energy from the MRI's (RF) field, potentially causing localized heating that poses a thermal risk to the patient. Third, the RF field may induce alternating currents within the lead, which can electrically stimulate nearby cardiac tissue, potentially leading to dangerous physiological effects [4]. Being the main topic of this document, translational motion and torque. Which remains under-characterized and that most of the research done in this subject shows the importance of translational forces in safety about common hospital objects that could potentially become projectiles while disregarding smaller. These are developed in Chapter 2.

In this document an explanation of the current state of research on this matter, the main elements involved in the interaction of temporary epicardial leads and MRI, especially in relation to mechanical interactions like translational force and torque are given. Current research, safety recommendations and methodologies to assess the magnetically induced forces involved between an MRI system and epicardial leads were explored. And the development of a in-lab methodology capable of testing these micro-Newton forces with precision later carried into clinical work was documented. All these, in order to clarify whether the static magnetic field in MRI poses a real risk to patients with retained leads.

To properly situate this work, the following section reviews pertinent literature on MRI QA devices, MRI safety principles, and previous studies to establish current state of the art regarding epicardial leads on MRI environment.

1.1 Literature Review

A large observational study by Schaller *et al.* . [18] evaluated 200 MRI examinations in 139 patients with a total of 243 abandoned leads. Their objective was to assess the safety of 1.5-T MRI in patients with at least one abandoned CIED lead and to evaluate any potentially harmful effects on the active CIED leads. The research found a low rate of adverse events (3%), with no serious complications such as sustained arrhythmias or device resets. The observed events were primarily transient, non-life-threatening changes in lead sensing parameters that resolved spontaneously. Notably, this safety profile held for thoracic and cardiac MRIs, where the theoretical risk is highest.

Complementing this, a retrospective study by Vuorinen *et al.* . [16] focused on 26 MRIs in 17 patients with epicardial leads. The study’s key finding was that MRIs performed on patients with ”modern” epicardial leads (implanted in 2000 or later) had no detectable adverse events. However, one definite and one possible adverse event were noted in patients with ”old” leads (implanted before 2000), suggesting that lead age and technology may be a factor in the risk profile.

From a safety standpoint, the static magnetic field (B_0) represents the main source of hazard and must be shielded properly. A huge number of diagnostic clinical scans are daily completed without incidents indicating that magnetic interactions with normal tissues are within the bounds of safety up to the highest fields now in use for MRI [4]. However, there had been incidents associated with ferromagnetic objects either implanted in the patient or located in the fringing field of the magnets [19, 20].

ASTM F2052-21 [5] offers a standardized technique for assessing magnetically induced displacement force on medical equipment in order to set safety standards for such situations. The methodology includes a non-magnetic pendulum apparatus with a protractor that allows for the a suspended subject to be pulled by the magnetic field. The characterization of the static magnetic field and its spacial gradient is required.

For this purpose, Ferreira *et al.* . [21], developed a custom-designed fixture with an AlphaLab Gaussmeter probe to map the spatial gradients inside a Siemens Magnetom Trio 3T MRI scanner. Measurements of the magnetic field vector components (X, Y, and Z) were collected along the Z-axis in 5cm increments, correctly capturing the static magnetic field and ensuring repeatability and showing a steady spatial gradient.

Furthermore, Woods *et al.* . [22], conducted a similar experiment, to assess the magnetically induced displacement force on eleven metal alloys commonly used in medical implants. For deflection force testing, nine specimens (in three nominal mass categories: 2, 20, and 200 g) of each alloy were tested in a Siemens MAGNETOM Trio 3T MR system. The main results regarding magnetically induced deflection force showed that for seven of the eleven tested alloys, the magnetic force was found to be less than the device weight in any current clinical 1.5 T, 3.0 T, and 7.0 T MR system. These safe alloys included various grades of CP titanium, Ti-6Al-7Nb, Ti-6Al-4V ELI, Ti-15Mo, Biodur 108, and Co-28Cr-6M. However, for the other alloys, including 316L stainless steel and 22Cr-13Ni-5Mn, the magnetically induced force was calculated to be greater than the device weight in certain high-field systems, such as 3.0 T and 7.0 T clinical scanners.

In terms of torque measurements, Woods *et al.* . [22] claimed that for medical instruments made from their list of alloys the magnetically induced torque will always be less than the worst case torque produced by gravity in a 3.0 T or less static magnetic field. And emphasized that “*No movement was observed for any of the magnetically induced torque specimens at 3.0 T.*” This, was probably measured using ASTM F2213-17 [23] “The Low Friction Surface Method” which estimates magnetically induced torque by placing the device on a low-friction, non-metallic, non-conductive surface near the MRI isocenter. The device is rotated stepwise, and any spontaneous alignment or rotation toward the static magnetic field is observed. If no movement occurs, the maximum possible torque is estimated from the device’s weight and the

surface’s friction coefficient outside the MR environment. ASTM F2213-17 states that, if rotation is observed, more precise tests (such as the Torsional Spring or Pulley Method) should be used instead which means this method relies on macroscopic motion and may not be able to detect μ Newton forces.

Another study by Seo *et al.* . [24], utilized “the Pulley Method” for the safety assessment of magnetically induced torques on active implantable medical devices (AIMD)s in 1.5 T and 3.0 T magnetic resonance imaging (MRI) environments. The apparatus, has a cylindrical shape and was fabricated entirely from acrylic (including the bearings) to minimize magnetic field resistance.

This methodology determines the maximum magnetically induced torque by fixing the AIMD onto a rotating plate placed at the MRI isocenter. A thread attached to the plate is connected to a force gauge located far enough to avoid interference with the magnetic field. As the plate rotates to align with field, the thread tension is measured. The torque is then calculated by multiplying the plate’s radius (at the AIMD’s position) by the measured force, after subtracting the plate’s frictional resistance.

The main results showed that the bone conduction implant consistently exhibited the highest average torque (9.460 Ncm at 1.5 T and 14.618 Ncm at 3.0 T), while the pacemaker showed the lowest (0.255 Ncm at 1.5 T and 0.668 Ncm at 3.0 T). AIMDs containing magnets (cochlear and bone conduction implants) generally showed significantly higher torques than devices without magnets (ICD and pacemaker).

Similarly, Stoianovici *et al.* . [25] showcases a variation of the ASTM F2213 Torsional Spring measurement technique, referred as the 2-string method. This method address the difficulty of measuring very low magnetically induced torques on lightweight, slender medical devices, such as needles. This approach utilizes two strings to suspend the device by its ends and creates a torsional spring mechanism that is frictionless.

The needle rotates due to magnetic torque, the torque is calculated analytically based on the measured rotation angle and compared to the weight of the object. The main results showed that the method successfully measured torques between 0.0001 Nm and 0.0018 Nm in the MRI environment. The maximum magnetically induced deflection angle remained below the 25° limit required by the ASTM F2213.

In contrast, the gradient and RF fields are primarily associated with peripheral nerve stimulation and tissue heating, respectively. These effects are relevant for patients with conductive leads or elongated implants that may act as antennas or current loops. Previous studies of Aboyewa, *et al.* . [8], seek to illustrate the interaction of leads with the RF field in terms of energy absorbed and heat produced.

The experiment was performed on a 3 T MRI scanner using a turbo spin echo pulse sequence. The pulse sequence was set to produce a whole-body Specific Absorption Rate (SAR) of approximately 2 W/kg and fluoroptic probes were used to measure the temperature rise. Several factors on induced heating were tested, including lead length (such as 6.33 cm, 12.65 cm, and 18.98 cm), lead type (bipolar vs. unipolar), orientation, and shape

It showed that lead length, type, and orientation significantly influence heating behavior. This study found that short leads (< 13 cm) posed no significant thermal hazard during MRI at a specific absorption rate (SAR) of 2 W/kg but the maximum temperature rise observed at the lead tips. For both lead types investigated, the amount of heat induced increased with increasing lead length. Notably, bipolar leads showed slightly higher heating than unipolar leads and the straight lead configuration produced the highest maximum temperature rise.

The overall significance of these studies is in establishing reliable experimental and computational frameworks for quantifying MRI-induced mechanical effects, particularly where conventional ASTM approaches fall short. .Stoianovici *et al.* . [25] solves the technical challenge of measuring very low magnetically induced torques

on slender, lightweight medical devices, such as needles, where conventional ASTM F2213 methods often become unreliable due to friction bias with their new methodology variant (The-2-string-method). Similarly, Seo *et al.* . [24] validates the pulley methodology for AIMDs. The modeling of clinically relevant allows done by Woods *et al.* . [22] shows experimental basis to rely on the listed alloys but requires further testing in terms of torque. While the modeling of the static magnetic field done by Ferreira *et al.* . [21] provided a practical way to accurately map the magnetic field and calculate its spacial gradient.

Finally, MRI safety protocols are grounded in a system of absolute and relative contraindications. Absolute contraindications include non-MR-conditional cardiac implantable electronic devices (CIEDs), metallic intraocular, and foreign bodies, among others. Relative contraindications consider implants such as stents, surgical clips, and programmable shunts that may be scanned under controlled conditions [26]. The safety classification of each device is guided by testing standards and certification data, with CIEDs representing a critical category due to their susceptibility to electromagnetic interactions [4, 27].

The American College of Radiology (ACR) Manual on MR Safety (2024) [28] outlines the current regulatory and procedural framework for MRI safety. Recommendations include structured zoned access control and interprofessional collaboration through clearly defined positions (MR Medical Director, MR Safety Officer, and MR Safety Expert) to avoid unscreened individuals or ferromagnetic objects from inadvertently entering the magnet room. Pre-scan screening, implant model and status verification, and continuous patient monitoring are still necessary for this safety infrastructure [26].

1.2 Scope of Research

The primary objective of this study is to experimentally characterize torque and translational magnetic forces acting on retained epicardial pacing leads near and inside the MRI bore, and to map how those forces evolve with distance from the isocenter.

The wider research question is: Do retained epicardial pacing leads experience torque or translational magnetic forces within an MRI environment that are clinically significant enough to justify current safety restrictions? A particular research question then follows: How to implement current methodologies to measure micro-Newtons on a clinic environment?

A lab methodology was first developed, working towards clinical environment and rigor addressing potential issues. Thus, this document aims to provide experimental evidence and addressing a gap in the literature. In the lab, investigation used a Polyvinyl Chloride (PVC) pendulum-based measurement setup following ASTM F2052 [5] for the displacement angle and a custom-made solenoid simulating a magnetic field similar to that found on a MRI scanner bore entrance at scale. Displacement in millimeters of a metallic rod exposed to the field were recorded to ultimately estimate the translational force.

A second setup for torque measurements was developed. A set of neodymium magnets would create the static field while a metallic rod would be fixed in a position perpendicular to the field with an attachment and be allowed to rotate, counter forces would then be measured out the attachment with a circuit and sensor of our own fabrication. A second iteration of this setup would include a Helmholtz coil to test on field variation.

These methods allow for high sensitivity in detecting small magnetic forces in a controlled environment, without the complications of patient variability or direct MRI scanner use. The pendulum approach provides a clear physical model that isolates

translational force components for precise characterization. The torque model allows to independently assess strength of force exerted.

In the following chapter, a thorough description of the experimental setup will be given. This includes the caveats found and their respective solutions with detailed documentation on the artifacts and tools created for the investigation.

The thesis consistently engages with the first of three interaction modes of concern between MRI and retained leads: mechanical forces, RF-induced heating, and induced currents. Particularly, the study cares about translational and rotational displacement due to static field gradients to clarify whether or not the movement involve signifies a risk to patients with temporal epicardial leads when an MRI scan is needed.

Measurements are interpreted in the context of magnetic susceptibility theory and compared to relevant thresholds from the literature [29, 5]. These techniques will help with more thorough evaluations of MRI safety in patients with retained temporary epicardial leads and set the stage for a future clinical adaptation of the protocol employing a 3T MRI scanner.

1.3 Summary

Epicardial leads have the problem of being broadly labeled MRI-unsafe under ASTM F2503. This classification origins from theoretical concerns over static magnetic field interactions (translational force and torque), RF-induced heating, and inadvertent electrical stimulation, yet it restricts MRI access for cardiac patients most in need of diagnostic imaging.

Hundreds of MRI tests using abandoned leads, including contemporary epicardial types, have shown no significant adverse events, indicating good safety profiles in large observational studies. There is still a significant gap in the accurate, micro-

Newton-scale characterization of mechanical forces on thin, lightweight epicardial leads, despite the fact that standardized testing (ASTM F2052, F2213 [5],[23]) and earlier research on implant alloys confirm low magnetic interactions in many materials. The methodology on this project follows that of the literature reviewed in this chapter. Featuring a static field mapping fixture, a pendulum for deflection angle determination, and a newly developed torque device that somewhat resembles a suspension method but uses a low friction bearing to enable rotation and a highly sensitive sensor to measure μ Newton forces, which is further described in Chapter ??.

Building on this foundation, Chapter 2 will provide essential background on the underlying physics of MRI scanners. It includes the generation and spatial distribution of static field, along with a broad introduction to gradient, and RF fields. Alongside physics, a detailed description of epicardial lead design, material composition, and implantation configurations is provided. Key concepts from electromagnetism (magnetic susceptibility, force, and torque on conductive elements) and mechanics (deflection, rotation, and tissue anchoring) will form the thematic backbone, enabling a rigorous analysis of lead-MRI interactions and informing the interpretation of experimental results in subsequent chapters.

Chapter 2

Theoretical Background

This chapter provides a review of the most important topics to know to fully understand the findings of this current work. It includes an introduction to MRI scanners, the physics behind them and, more importantly, the possible interactions of the system with matter. Moreover, an emphasis on the static magnetic field of a MRI scanner and mechanical interactions with epicardial leads is included. A brief discussion on epicardial leads design is included followed by implications for the clinic and the lab setting.

Most of this chapter content is based on physics and imaging from well known textbooks, including Griffith's Introduction to Electrodynamics [30], and Bushberg's Physics of Medical Imaging [3].

2.1 Epicardial Leads

CIED systems refer to cardiac implantable electronic devices, including pacemakers, implantable cardioverter-defibrillators (ICD)s, cardiac resynchronization therapy (CRT) devices, and implanted rhythm monitors. These are advanced medical devices used to monitor and regulate heart rhythm disorders and leads are a component of CIED systems.

In practice, leads are a type of very thin, ~ 4 to 5 F (French gauge, meaning 1.3 to 1.7 mm), electrical wire that connects a pulse generator to the heart. Figure 2.1 shows the two types of leads that are used, epicardial leads; which are implanted directly on the epicardium (the outer surface of the heart) and Transvenous leads, inserted through the veins and positioned inside the heart's chambers, making contact with the endocardium (the inner lining of the heart). All these are prone to complications like fracture, infection, venous obstruction (transvenous only), and require complex extraction procedures [31, 32].

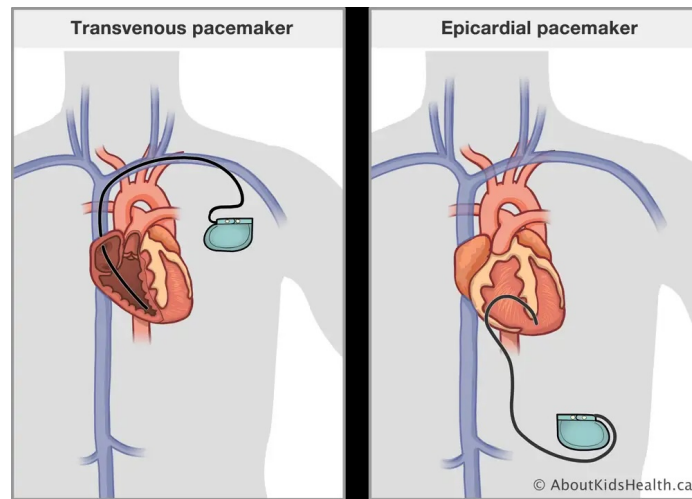
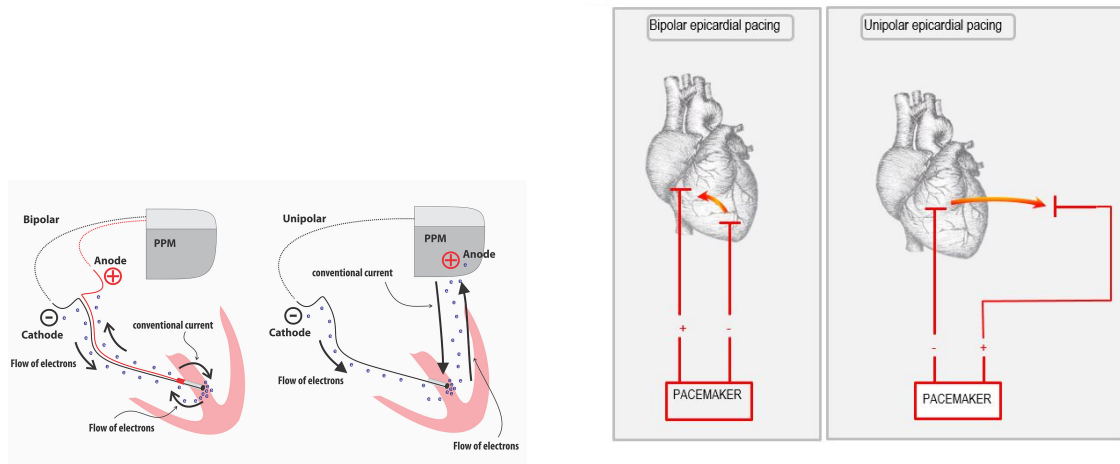


Figure 2.1: Comparison of transvenous vs epicardial lead placement in pacemakers (adapted from [1]).

The position of leads is an important distinction. Clinically, these leads are indicated for different populations and diagnostics therefore carry a variety of management decisions (leaving vs extracting) and different complications (percutaneous lead extraction or transvenous lead extraction). [17, 33] Physically, different contact with tissue means mechanic interaction, an example is that transvenous leads may be cooled by blood affecting its overall heating capacity. Or that any MRI-induced displacement force or vibration translates directly to mechanical stress for epicardial leads, being sutured directly to the epicardial tissue. While tension on transvenous ones may be somewhat dampened by surrounding blood and vessel walls.

Within epicardial leads, design may vary, examples are from lead polarity and body structure. On unipolar leads, a single conductor and one electrode (the cathode) are in contact with the heart, while bipolar leads contain two conductors that connect to two electrodes on the lead itself, a tip electrode (cathode) and a ring electrode (anode). In figure 2.2 the comparison of these types is shown, on the left side (2.2(a)) depicts the current flow for each case while the right side (2.2(b)) shows from outside the position of the anode relative to the heart.

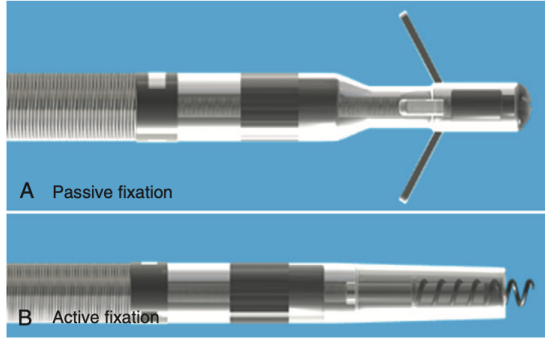


(a) Unipolar vs. bipolar epicardial lead configuration.

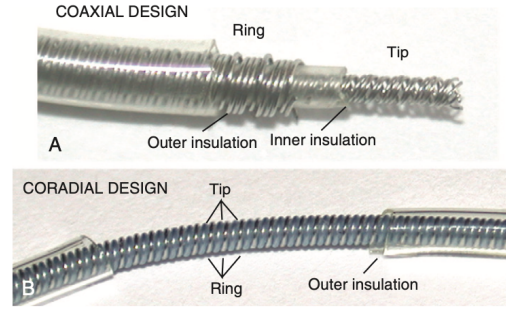
(b) Relative electrode positioning of unipolar and bipolar epicardial leads.

Figure 2.2: Comparison of unipolar and bipolar epicardial pacing leads. (a) Illustrates the distinction in current flow pathways using a schematic of the heart, adapted from [34, 35]. (b) Depicts the relative anode and cathode positions in unipolar and bipolar epicardial leads, adapted from [36, 35].

Bipolar leads then, have a body structure design variation. As shown in Fig. 2.3, coaxial design features an inner coil conductor nested within an outer coil conductor, with each wrapped in a separate layer of insulation while coradial design consists of a single coil containing two parallel, intertwined conductor strands, each individually insulated [35]. Furthermore, epicardial leads, when abandoned, encompass this diversity in design and composition as all components of the leads have different materials.



(a) Lead fixation mechanisms.



(b) Coaxial and coradial lead body design.

Figure 2.3: Lead fixation and body design variations in pacing leads. (a) Passive-fixation leads rely on flexible tines that anchor within myocardial trabeculae, while active-fixation leads employ a retractable helix that screws into the myocardium. (b) Coaxial designs use two nested coil conductors with separate insulation layers, whereas coradial designs incorporate parallel, intertwined conductors within a shared lumen. Adapted from [35].

The several components of an epicardial lead are made of different materials chosen for their biocompatibility, durability, and electrical properties. Electrodes are made from inert materials to minimize corrosion and inflammatory reactions over the long term. These include titanium, platinum-iridium alloys, Elgiloy (an alloy of cobalt, iron, chromium, and other metals), and vitreous carbon. Similarly, conductors which carry the electrical signals between the pulse generator and the electrode are commonly made out of MP-35N (alloy of nickel, cobalt, chromium, and molybdenum.) Then, insulation, a protective outer coating of the lead that prevents short-circuiting made of different polymers like Silicone Rubber (Polysiloxane), Polyurethane, Fluoropolymers.

Finally, epicardial leads are affixed directly to the heart's outer surface during a surgical procedure in mainly two ways, as shown in figure 2.3(b). Suturing and Screwing, meaning the electrode is designed to be sutured directly onto the epicardium. While screwing refers to an active-fixation helix (screw), typically made of a platinum-iridium alloy or a similar metal, screwed directly into the myocardium [35]

From figure 2.3 one can see some of the challenges when retracting the epicardial lead, for this practical reason they are commonly abandoned after its purpose has been fulfilled. An abandoned epicardial lead could measure full-length or fragments after being left behind. Some fragments are described as long as 5cm, being 2cm an average length. Further, longer leads are reported up to 30 cm or couple fragments adding to about 21 cm. In diameter size may be compared to standard transvenous pacing leads, that of about half a millimeter to 1.5 mm. And are typically narrower than standard transvenous leads ($\sim 5F$, $\simeq 1.7$ mm outer diameter), often measuring between 0.5 mm and 1.5 mm.”

The physical length of the wire can impact the resulting interactions with the magnetic field, as more of the material in the human body more of it can interact. Table 2.1 summarizes the materials found on a epicardial lead and relates to its magnetic susceptibility.

Table 2.1: Qualitative magnetic susceptibility of common epicardial lead materials and their relevance for MRI safety. [12, 37, 38, 39]

Component	Material	Magnetic susceptibility	Relevance for MRI safety
Electrode	Titanium	Paramagnetic	Low; minimal torque/force in MRI
Electrode	Platinum-Iridium alloy	Paramagnetic	Low; generally MRI safe
Electrode	Elgiloy (Co-Cr-Fe-Ni alloy)	Paramagnetic	Moderate; may experience slight translational force
Electrode	Vitreous carbon	Diamagnetic	Minimal; very low MRI interaction
Conductor	MP-35N	Weakly paramagnetic	Low; minimal contribution to MRI forces
Insulation	Silicone rubber (polysiloxane)	Diamagnetic	Negligible; non-conductive and non-magnetic
Insulation	Polyurethane	Diamagnetic	Negligible; non-magnetic
Insulation	Fluoropolymer (PTFE)	Diamagnetic	Negligible; MRI inert

Current institutional recommendations regarding MRI in patients with retained epicardial leads remain conservative despite evolving evidence. Major guideline bodies, including the Heart Rhythm Society (HRS), the European Society of Cardiology (ESC), and the Centers for Medicare & Medicaid Services (CMS), allow MRI in patients with MRI-conditional or certain legacy cardiac implantable electronic devices (CIEDs), but continue to restrict or discourage imaging in patients with abandoned or epicardial leads [16].

Specifically, at the time of the HRS 2017 consensus and the 2021 ESC pacing guidelines noted that there was insufficient data to comment on the safety of MRI with an abandoned cardiac lead. And that if a patient had an MRI-conditional system along with an abandoned lead, lead extender, lead remnants, fractured lead(s), or surgically implanted epicardial leads, the system was considered non-MRI-conditional. Leading to a Case-by-Case consideration for whether to image or not and in which modality. CMS similarly excludes abandoned or epicardial leads from coverage due to perceived insufficient safety data, even though large observational studies indicate that modern epicardial leads (implanted after 2000) may undergo MRI safely under carefully monitored conditions [16, 18].

This cautious stance of institution remark the need for experimental characterization of mechanical interactions between retained epicardial leads and MRI fields. Thus, providing a rationale for the present study. By quantifying these forces under controlled laboratory conditions, the work shown in this document directly addresses whether current institutional restrictions are justified and lays the groundwork for evidence-based updates to clinical safety policies.

2.2 Magnetic Resonance Imaging

MRI or Magnetic Resonance Imaging is a medical imaging modality that uses magnetic fields and radiofrequency (RF) energy to produce images, rather than ionizing radiation like X-rays making it inherent safe. Moreover, MRI machines are capable of achieving high contrast sensitivity to soft tissue differences because of the reliance on micromagnetic properties of the tissue and its inherent differences leveraging on Proton Density (PD) and the relaxation phenomena.

MRI, as the name suggests, leverages the magnetic properties of matter (its atoms) to capture information by applying RF energy at the resonance frequency and detects

the re-emitted signals to ultimately create an image. In fact, MRI was originally named Nuclear Magnetic Resonance (NMR), because it involves the spectroscopic study of the magnetic properties of the nucleus of the atom as it resonates and aligns to the magnetic field changes.

It is important, then, to understand these properties and how they are characterized. On the scale of elementary particles, these effects involve interactions with protons and neutrons within the atomic nucleus due to their nuclear spin and charge distribution. Hydrogen (specifically its proton, 1H) is the most utilized element in clinical MRI because it has the largest magnetic moment and is highly abundant in biological tissues, primarily in water and fat.

Normally, these unbound hydrogen atoms have a random orientation of their nuclear magnetic moments (protons) due to thermal energy. When placed in a strong static magnetic field (B_0), these nuclear magnetic moments align in parallel (same direction of B_0 , low-energy state) and antiparallel (opposite direction of B_0 , high-energy state) directions. This is illustrated in Figure 2.4, notice that the distribution of protons into these two states is influenced by thermal energy and a slight majority align in the low-energy, parallel direction, resulting in an observable net magnetic moment (red arrow) usually denoted as equilibrium magnetization (M_0).

In addition to aligning with B_0 , the protons also experience precession, much like the Earth's rotation around its tilted axis or a spinning top under gravity. Depicted in Figure 2.5, the motion happens around the axis of the applied magnetic field due to the perpendicular torque they experience. This precession occurs at an angular frequency (ω_0) known as the Larmor frequency, which is directly proportional to the magnetic field strength (B_0), as described by the Larmor equation (2.1):

$$\omega_0 = \gamma B_0. \tag{2.1}$$

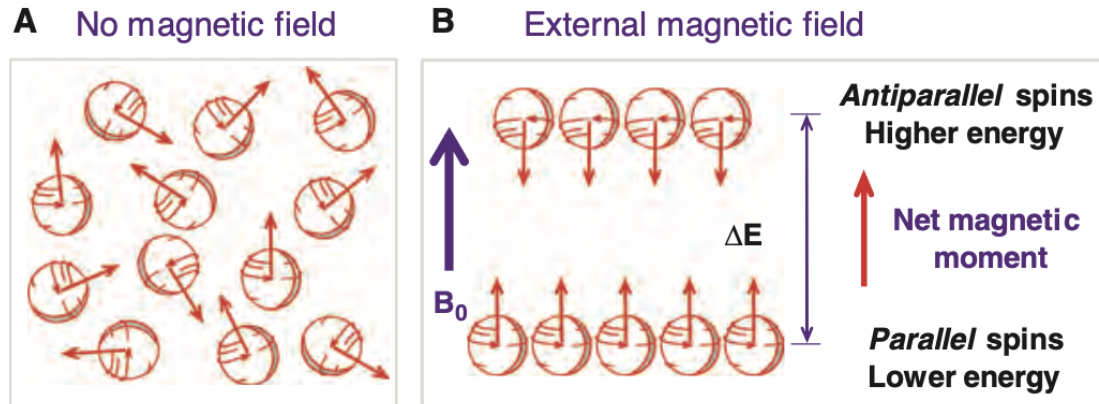


Figure 2.4: Schematic illustration of how hydrogen protons (^1H) behave in the presence of a strong static magnetic field (B_0). A: In the absence of a field, proton magnetic moments are randomly oriented, producing no net magnetization. B: When exposed to B_0 , protons align either parallel or antiparallel to the field. From Bushberg, J.T., *The Essential Physics of Medical Imaging* (2011) [3].

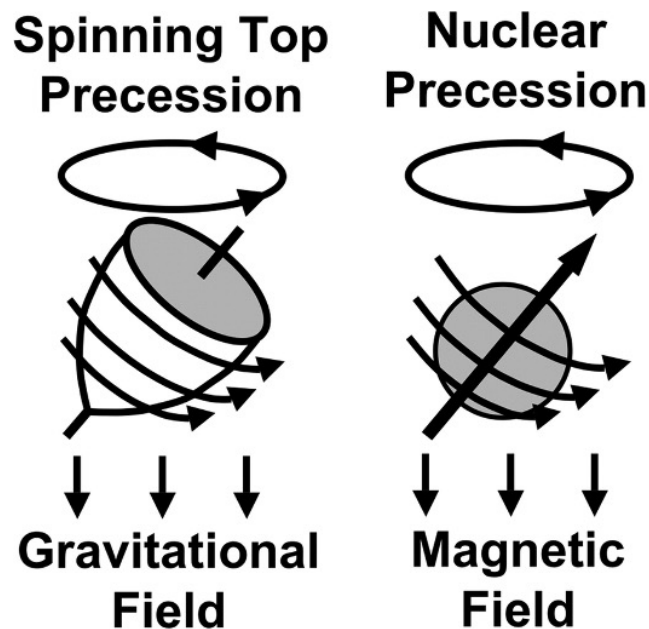


Figure 2.5: Illustration of proton precession under a static magnetic field B_0 compared to that of a spinning top under gravity. Adapted from Pooley, R. A., *Fundamental physics of MR imaging*, (2005) [2].

An RF energy pulse (B_1) is then applied, synchronized to the Larmor frequency of protons (about 42.58 MHz per tesla), magnetizing the protons and tipping the net magnetization (M) away from the longitudinal direction (M_z). This pulse provides

just enough energy for some protons to move into a higher energy state, also called RF excitation. In doing so, it also causes the magnetic moments of many protons to become synchronized, or phase-coherent. They start to precess together in the transverse plane, creating what is called transverse magnetization (M_{xy}).

The instant the RF pulse is turned off, the relaxation process starts, these precessing protons start to lose that synchronization due to slight variations in their local magnetic environments. This causes the transverse magnetization signal to gradually fade over time. The oscillations of this signal are detected by the same RF coils that sent them. The strength and rate of decay of this signal are dependent on the tissue type and are characterized by two time constants: $T1$ and $T2$.

First, the signal in the transverse plane, (M_{xy}), decreases as the protons desynchronize, this is described by $T2$ (Transverse Relaxation). And the process by which the system regains equilibrium is described by $T1$ (Longitudinal Relaxation). The longitudinal magnetization (M_z) returns, and the particles realign with the main magnetic field (B_0). The final MRI image's depiction of a tissue's brightness or darkness is determined by these relaxation processes taken together. For instance, on $T1$ -weighted images, tissues with short $T1$ times (such as fat) seem bright, whereas on $T2$ -weighted images, tissues with longer $T2$ periods (such as cerebrospinal fluid) appear bright, see Figure 2.6.

However, this covers one a single signal from a particular spot. To make an image rather, the MRI system must know where each signal is coming from. This is achieved using small, precisely controlled variations in the magnetic field known as gradient fields. These gradient coils, superimposed on B_0 , make the magnetic field slightly stronger or weaker depending on position, causing the Larmor frequency of the protons to vary with location.

Three gradient systems work together to map space. First, Slice Select Gradient (SSG) is applied during excitation so that only protons in a particular “slice” of the

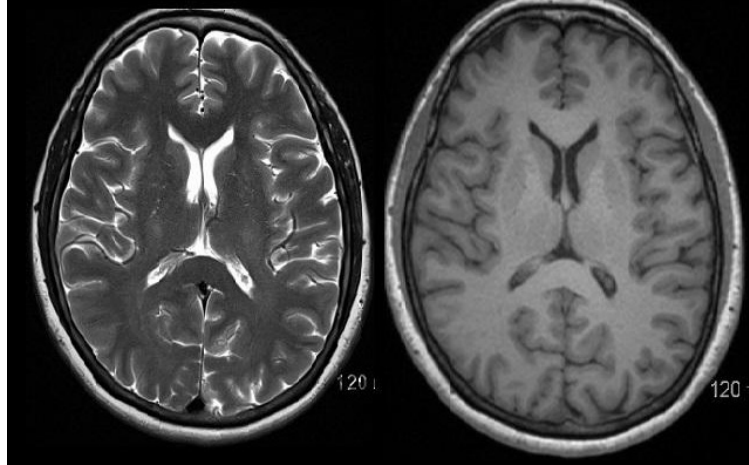


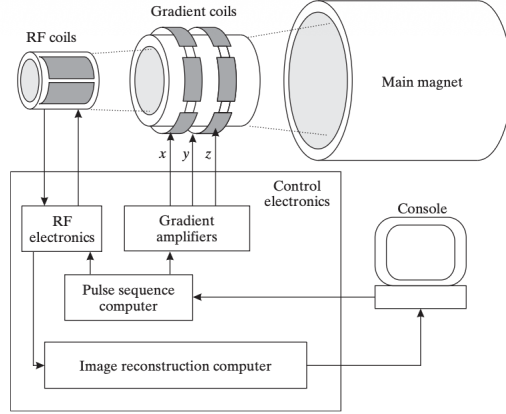
Figure 2.6: Comparison between T1- and T2-weighted brain MRI images of the same subject. The left image shows a T2-weighted scan, where cerebrospinal fluid (CSF) appears bright, and the right image shows a T1-weighted scan, where fat and white matter appear bright. Adapted from [40].

body resonate at the RF pulse frequency. This defines the plane of imaging. Then, Phase Encoding Gradient (PEG), briefly changes the precession speed of protons in one direction, giving each row of tissue a unique phase “signature”. Finally, Frequency Encoding Gradient (FEG) is applied during signal readout so that protons along another axis precess at slightly different frequencies.

The resulting signals are stored in a matrix called k-space. Each line in k-space represents one measurement with a specific gradient configuration. When all necessary lines are filled, the Fourier Transform reconstructs the data into an image, where each pixel corresponds to a small volume element (voxel) of tissue.

So far we have highlighted three of the five main components in the image formation process. That is, Static Field, RF pulses and Field gradients. Figure 2.7(a) shows all these components and Figure 2.7(b) provides an example of a actual, clinically relevant MRI scanner like the one used at CHI Health Creighton University Medical Center - Bergan Mercy. Each of these pieces are accompanied by interaction complications if a foreign object lays around the test space. For example, as explained in section 1.1 heat and electrical interactions are a concern of RF fields or Gradient

Coils [8, 18]. For mechanical interactions, the interest falls under the mechanisms in which the static magnetic field could possibly interact with wire or thin cylinder like devices. In the next section a review of an MRI scanner components is given, followed by a complete explanation on the main magnet as it is the focus of this document.



(a) Diagram illustrating the main five components of an MRI scanner.

(b) Siemens 3T Magnetom Trio clinical MRI system.

Figure 2.7: MRI scanner schematic and a clinical MRI system. (a) Simplified MRI scanner design highlighting the main magnet, gradient coils, and RF system, Control electronics, and Console. Adapted from Prince [41]. (b) Siemens Magnetom Trio 3T system, like at CUMC Bergan Mercy Hospital. As a modern whole-body MRI scanners, adapted from [42].

2.2.1 Scanner Design and Subsystems

As mentioned above, after the main static field (B_0) has been established and shimmed to high homogeneity the gradient coils introduce controlled spatial variation in the magnetic field for encoding localizing signals generated during MR. Gradient coils, are wire conductors located within the main bore of the magnet with typical strength values from 1 to 50 mT/m and correspond to the logical coordinate axes: x, y, and z which relate to SSG, PEG, and FEG. They operate by adding to or

subtracting from the magnitude of the main field (B_0), but they do not change the direction of the magnetic field [3].

These gradients must switch rapidly and with large current flows, typical slew rate ranges from 5 to 250 mT/m/ms, with currents on the order of 100–200 A [3, 41], to accomplish encoding. For this reason, they introduce hazard mechanisms beyond magnetic fields. For example, when the gradient currents are repeatedly turned on and off, large forces (Lorentz forces) are applied to the coils in the strong magnetic field, leading to significant vibration. That, is the source of the loud "knocking" sounds during MR scans. Noise levels in modern MRI systems can easily exceed 130 dB and temporary hearing loss was reported in early studies [43]. Consequently, hearing protection is now considered mandatory during MRI procedures.

Moreover, the rapid temporal variation of MRI gradient fields induces electric fields within conductive tissues via Faraday's law. Although these fields are much weaker than those associated with RF transmission in terms of tissue heating the time-varying electric potentials can depolarize excitable membranes, producing peripheral nerve stimulation (PNS) or, in extreme cases, involuntary muscle twitching [43, 44]. The effect scales with both the rate of change of the magnetic field and the spatial extent of the conductive path through tissue, so even though the absolute magnetic field perturbation from gradients (10–60 mT/m, or less than 1% of B_0) is small, the rapid temporal ramping can induce voltages large enough to reach neural activation thresholds [45]. Typical clinical systems, limited by FDA guidelines to ~ 40 T/s, remain below the threshold for cardiac excitation, but research systems capable of 300 mT/m gradients and higher slew rates approach that regime, posing theoretical cardiac stimulation risks [3, 41, 43].

RF coils and heating

The control and computing subsystems coordinate every stage of image acquisition and reconstruction. The host computer and operator console provide the inter-

face through which the technologist selects and supervises scanning protocols, while the pulse programming and measurement control systems synchronize gradient and RF activity [3]. Supporting units such as the digitizer, image processor, and compute engine translate the raw analog signals into interpretable digital images through Fourier-based reconstruction algorithms [41]; Finally, the dedicated power supplies, patient table mechanics, and electromagnetic shielding ensure stable operation, patient positioning, and noise minimization.

2.2.2 The Static Field

Magnets are at the core of MR systems, and their performance is judged by field strength, temporal stability, and homogeneity. The strength of a magnetic field, also known as magnetic flux density (B), is measured in Tesla (T). An older, alternative unit is the gauss (G), where $1 \text{ T} = 10,000 \text{ G}$. For perspective, Earth’s magnetic field is significantly weaker, around 0.00005 T (0.05 mT or $0.5\text{-}1.0 \text{ G}$).

Magnetic fields can be induced by moving charges in a wire, where the direction depends on the charge’s sign and direction, described by the ”right-hand rule”. Coiling a current-carrying wire, creating a solenoid or electromagnet, superimposes these magnetic fields, significantly augmenting the field strength inside the coil and causing a rapid falloff outside. Equation 2.2 shows how the amplitude of the current and the number of coil turns determine the overall magnitude of the magnetic field strength for an ideal solenoid and the shape of such field is depicted in figure 2.8.

$$B = \mu_0 nI. \tag{2.2}$$

Where B is the magnetic field, μ_0 is the vacuum permeability, n is the number of turns and I is the current. Common MRI are typically made of superconducting wire-wrapped cylinders. These wires are made from niobium-titanium alloys and show

no resistance to electric current when maintained at extremely low temperatures. Notice that on Figure 2.8 produced magnetic field is parallel to the cylinder's long axis, usually horizontal (along the patient's cranial-caudal axis in clinical systems).

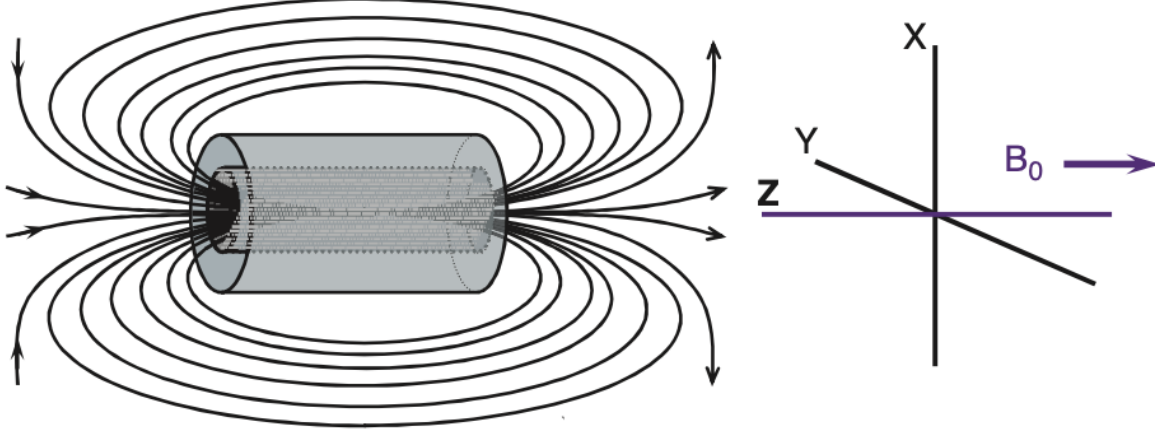


Figure 2.8: Ideal magnetic field lines produced by a solenoid coil configuration. The right side shows the typical axis labeling for an MRI scanner. Adapted from [3].

A more accurate depiction of the magnetic field can be obtained by modeling a finite solenoid of length L and radius R using a formula derived from Biot–Savart law with $B = 3\text{T}$ from equation 2.2 as follows:

$$B(z) = \frac{\mu_0 n I}{2} \left[\frac{z + \frac{L}{2}}{\sqrt{R^2 + \left(z + \frac{L}{2}\right)^2}} - \frac{z - \frac{L}{2}}{\sqrt{R^2 + \left(z - \frac{L}{2}\right)^2}} \right]. \quad (2.3)$$

where μ_0 is the permeability of free space. The first term in the brackets represents the contribution from the far end of the solenoid, and the second term represents the contribution from the near end.

This formula provides the axial field along the solenoid, with the maximum occurring at the center ($z = 0$) and decreasing smoothly toward the ends. Then, calculating the gradient and finding its maximum value, the bore entrance can be located [5, 21]. Figure 2.9 showcases this model, it features the magnetic field (B_0) along the z -axis, the origin $z = 0$ corresponds to the entrance of the bore. The field is normalized to

a maximum of $B_0 = 3 \text{ T}$, with no shielding or distortion applied, the solenoid sizes used on the simulation were 1m length by 0.6m diameter.

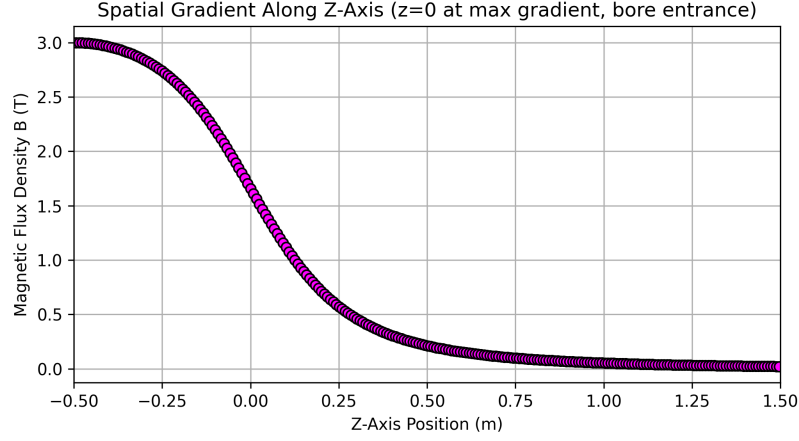


Figure 2.9: Expected 1D axial magnetic field B_z for an ideal solenoid, mimicking an MRI scanner.

From the same model a 3D render with slices of the field can also be constructed. Figure 2.10 is the resulting plot with a reference cylinder superimposed to indicate the bore space.

Let us then follow the physics of a static magnetic field and its interaction with matter with electromagnetism theory. Any material placed in a magnetic field develops a magnetization, $\mathbf{M}$, defined as the net magnetic dipole moment per unit volume:

$$\mathbf{M} \equiv \frac{\mathbf{m}}{V}. \quad (2.4)$$

where $\mathbf{m}$ is the sum of the microscopic magnetic dipoles within the volume V [30]. In practice, the microscopic magnetic field inside a material fluctuates rapidly due to atomic and electronic motion. To describe macroscopic behavior, we use an average field over many atoms, and $\mathbf{M}$ represents this “smoothed” magnetization.

To simplify the analysis of materials in magnetic fields, physicists define the auxiliary magnetic field $\mathbf{H}$:

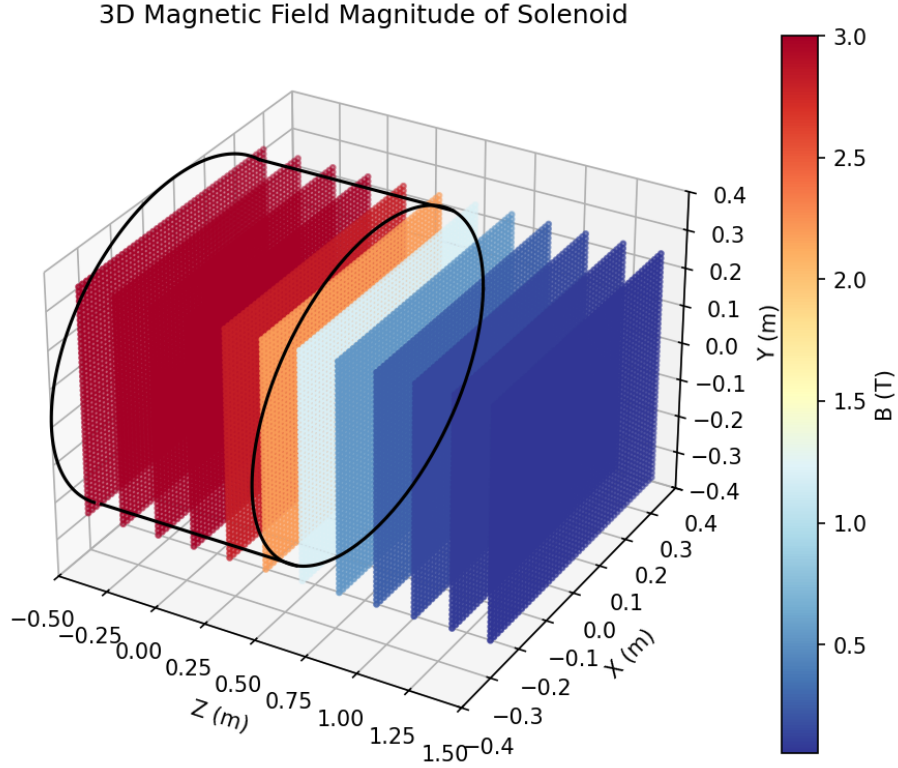


Figure 2.10: 3D rendering of the axial magnetic field of a 3 T MRI solenoid.

$$\mathbf{H} \equiv \frac{1}{\mu_0} \mathbf{B} - \mathbf{M}. \quad (2.5)$$

where $\mathbf{B}$ is the macroscopic magnetic flux density and μ_0 is the permeability of free space. $\mathbf{H}$ accounts for the part of the magnetic field produced by free currents, separating it from the bound currents induced by magnetization [30]. This distinction becomes useful when analyzing mechanical effects on magnetic dipoles.

Different materials respond differently to an applied field. As magnetization is also directly proportional to the field strength $\mathbf{H}$:

$$\mathbf{M} = \chi_m \mathbf{H}. \quad (2.6)$$

The degree to which a material becomes magnetized is quantified by its magnetic susceptibility, χ , which is typically small in biological tissues ($|\Delta\chi| < 0.01$). Dia-

magnetic materials, such as water, calcium, and most organic compounds, develop a magnetization opposite to the applied B-field due to paired electrons. Paramagnetic substances, like molecular oxygen, deoxyhemoglobin, or gadolinium-based MRI contrast agents, align weakly with the field, enhancing the local magnetic flux. Ferromagnetic materials, by contrast, exhibit strong self-sustained magnetization but are generally avoided in MRI due to safety hazards. Figure 2.11 shows a spectrum of magnetic susceptibilities with a zoom in for the MRI compatibility obscure section.

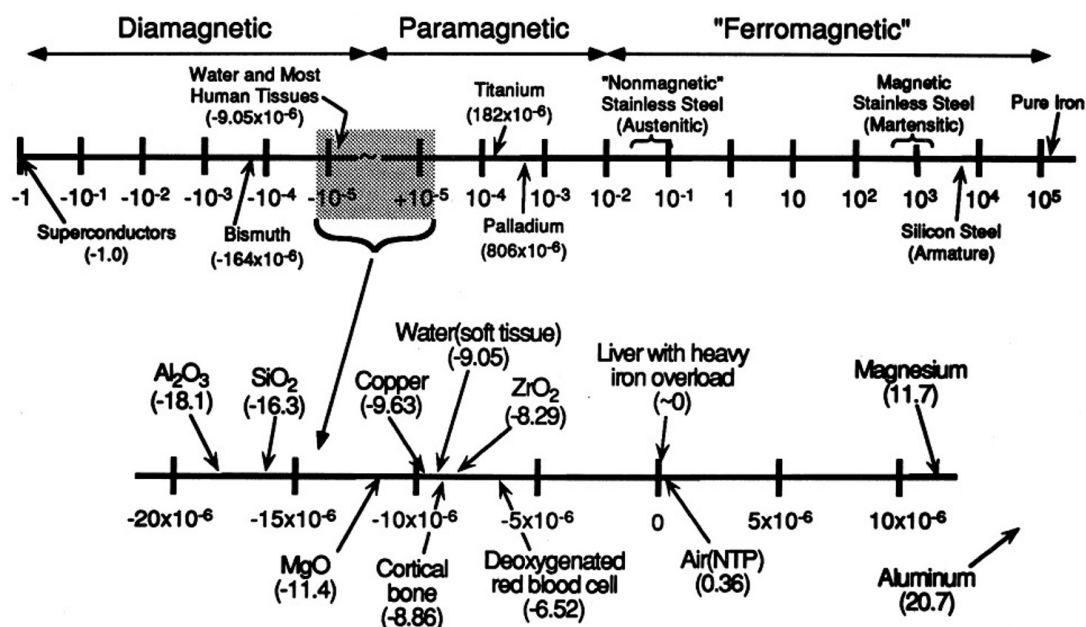


Figure 2.11: Spectrum of magnetic susceptibilities. The upper diagram uses a logarithmic scale to indicate the full range of observed magnetic susceptibility values: it extends from -1.0 for superconductors to 100,000 for soft ferromagnetic materials. The bottom diagram uses a linear scale (in ppm) to indicate the properties of some materials with 20 ppm. The susceptibilities of most human tissues are in the range from 7.0 to 11.0 ppm. Adapted from Schenck [4].

For materials that obey Eq. 2.6, called *linear media*, the magnetic flux density can be expressed as

$$\mathbf{B} = \mu_0(1 + \chi_m)\mathbf{H}, \quad (2.7a)$$

$$\mathbf{B} \approx \mu_0\mathbf{H}, \quad (|\chi_m| \ll 1). \quad (2.7b)$$

where χ_m is the magnetic susceptibility of the material. We can define the *permeability* of the material as $\mu \equiv \mu_0(1 + \chi_m)$, so that the relationship between $\mathbf{B}$ and $\mathbf{H}$ simplifies to $\mathbf{B} = \mu\mathbf{H}$. Moreover for paramagnetic materials which susceptibility is tiny ($|\chi_m| \ll 1$) $\mathbf{B}$ can approximate to $\mathbf{B} \approx \mu_0\mathbf{H}$.

This formulation shows that in linear media, the magnetic flux density is directly proportional to the applied field strength, statement that will come in handy when calculating for forces and torque in the next section. In the particular case of magnetic dipoles $\mathbf{m}$, mechanical interactions with a magnetic field such as forces and torques the following expressions are the foundation from general physics.

$$\mathbf{F} = (\mathbf{m} \cdot \nabla)\mathbf{B}. \quad (2.8)$$

and a torque:

$$\boldsymbol{\tau} = \mathbf{m} \times \mathbf{B}. \quad (2.9)$$

where $\mathbf{m}$ is the magnetic dipole moment and B is the magnetic field. From these two equations we can solve for more specific physical setups. For translational force the equation (2.8) is developed following ASTM F2052-15, as same methodology was employed. While equation (2.9) for torque is dependent to the particular procedure used on the lab, and will be developed in more detail based on our own experimental setup. For paramagnetic materials (case of eq. 2.7b), we can estimate the magnetic

moment as the product of the magnetic susceptibility, magnitude of the field strength, and volume of the material.

$$m_0 = \frac{1}{\mu_0} \chi_m B_0 V \quad (2.10)$$

2.2.3 Translational Forces induced by the Static Field

Consider a simple pendulum with a metallic object at its end. This object is free to swing along the z-axis of an MRI. A static magnetic field along that same axis is then applied. The pendulum is then pulled towards the magnet with a force F_m until it reaches equilibrium with its weight mg . Figure 2.12 shows a complete free body diagram, a simpler version excludes Θ and assumes point particle at the end of the pendulum.

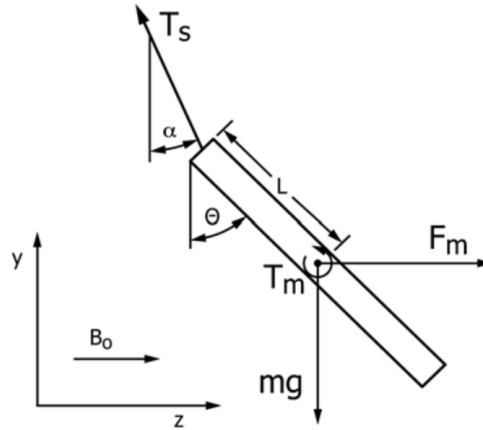


FIG. X2.1 Free Body Diagram of Device in Magnetic Field

Figure 2.12: Free-body diagram of device in magnetic field (adapted from ASTM F2052-15 [5], Appendix X2).

The magnetic force F_m is described from equation (2.8) and using equation (2.4) and (2.6) we can write:

$$\begin{aligned}
F_m &= MV\nabla B, \\
&= \chi_m HV\nabla B.
\end{aligned}$$

which is the the claim from before (see eq. 2.10). Finally, with equation (2.7b) the magnetic force in a single dimension is given by:

$$F_m = \frac{\chi_m V}{\mu_0} B \nabla B \quad \Rightarrow \quad \boxed{F_m = \frac{\chi_m V}{\mu_0} B_0 \frac{dB_0}{dz}} \quad (2.11)$$

then following ASTM F2052-15, the translational force exerted on a device due to a static magnetic field spatial gradient can be evaluated through angular deflection. In the pendulum, test subject is suspended in equilibrium, and the balance of forces leads to

$$\frac{F_m}{F_g} = \frac{F_m}{mg} = \tan(\alpha). \quad (2.12)$$

where α is the deflection angle from vertical, F_m is the magnetically induced translational force, m is the mass of the object, and g is the gravitational acceleration.

This is also derived from Newtonian force balancing (see Fig 2.12), and is valid even in cases when the magnetic force does not operate exactly at the center of mass. As a result, the approach is reliable for evaluating a variety of devices. The formulation of equation (2.12 explains why the measured deflection angle α (and thus F_m) depends not on χ or B_0 independently, but on their product. Practically, this insight justifies the lab methodology. Although magnets in our lab , a solenoid or neodymium magnets, generate small magnetic fields (only about 20 to 300 Gauss), we are ought to use a ferromagnetic sample, allowing us to replicate clinically relevant

force-to-weight ratios (F_m/F_g). An estimation for the expected χ_m can be found on appendix B.1 and summarized on table 2.2 below.

According to ASTM F2052 [5], a deflection below 45° corresponds to $F_m/(F_g) < 1$, meaning the magnetic force is less than the object's weight, a conservative reference point for safety, though not a strict acceptance criterion [22]. In contrast, a nonmagnetic stainless steel wire ($\chi \approx 10^{-4}$) might show $F_m/(mg) \approx 0.3$, whereas ferromagnetic metals like nickel can exceed $F_m/(F_g) > 2 \times 10^4$. That is more than enough to become dangerous projectiles unless restrained [8, 22, 43].

In our lab, we aimed to determine the magnetic susceptibility χ_m of the test sample. Using Eqs. (2.11) and (2.12), it can be expressed as

$$\chi = \frac{\mu_0 mg \tan(\alpha)}{V B_0 \frac{dB_0}{dz}}. \quad (2.13a)$$

$$\frac{\chi}{\rho} = \mu_0 g \frac{\tan(\alpha)}{B_0 \frac{dB_0}{dz}}. \quad (2.13b)$$

this provides a direct estimation of magnetic susceptibility from angular deflection measurements, provided the field and its gradient are known. For experimental purposes, a plot of the deflection angle versus the product of field strength and field gradient allows one to extract χ_m from the slope of a linear fit, following

$$\boxed{\tan(\alpha) = \frac{\chi_m}{\rho \mu_0 g} \cdot B_0 \frac{dB_0}{dz}} \quad (2.14)$$

In general, these forces are described as a function of the static field strength (B_0), its spatial gradient (∇B_0), and the material properties such as magnetic susceptibility (χ) and density (ρ). In practice the spatial gradient can be obtained by taking the finite difference in to very close points on a field strength map along the z-axis

$$\frac{dB}{dz} \approx \frac{B_2 - B_1}{z_2 - z_1} \quad (2.15)$$

where B_1 and B_2 represent the measured magnetic flux densities at positions displaced symmetrically from the location of interest. Using a 1-mm separation ($z_2 - z_1$) keeps the approximation in the linear regime while remaining resolvable on the field map. With this gradient in hand, the expected behavior is that the magnetic pull on high susceptibility materials toward regions of higher field strength, like near the bore entrance of the scanner where the spatial gradient is highest [3, 8, 43].

From equation (2.14), the measurable magnetic deflection depends on the ratio between the mechanical term and the magnetic forcing term. To reproduce the same deflection, 1 cm, at laboratory field strengths much smaller than those in MRI systems, the product must be preserved. Because $B_0 \nabla B_0 \propto B_0^2$ for a geometrically similar field, scaling the field down by a factor of 10^2 requires increasing the sample's effective susceptibility by roughly 10^4 to maintain the same mechanical response.

Table 2.2 summarizes the scaling. For typical solenoids producing 30–300 G the factor of χ_m/ρ spans from approximately one hundred-thousandth to one-thousandth of that found in ferromagnetic materials. Consequently, to appropriately mimic translational forces under these laboratory conditions, a realistic value of χ_m for a ferromagnetic substitute should lie between one and ten.

Table 2.2: Order-of-magnitude summary of laboratory scaling for required susceptibility. Values shown are representative from a scaled solenoid model in figure 2.9.

B_0 (T)	$B \cdot \nabla B$ (T ² /m)	χ/ρ	χ_m
3.0×10^{-3}	1.707×10^{-5}	7.21×10^{-3}	5.05×10^1
3.0×10^{-2}	1.707×10^{-3}	7.21×10^{-5}	5.05×10^{-1}
3.0×10^0	1.707×10^1	7.21×10^{-9}	5.05×10^{-5}

2.2.4 Torque induced by the Static Field and Torque Transfer

Similar to translational forces, let us start by explaining the physical system. Consider a metallic rigid cylinder shaped subject (a wire or rod) free to rotate, in a low friction surface, and align itself to a static magnetic field (B_0). The cylinder experiences a torque and has magnetic dipole moment according to eq. (2.9). For a thin, linear paramagnetic material, the induced magnetic moment depends on its orientation relative to the static magnetic field. It can be expressed as

$$\mathbf{m} = m_0 \cos(\theta), \quad (2.16)$$

where m_0 is the maximum induced dipole moment, which occurs when the material's axis is aligned parallel to the B_0 field ($\theta = 0^\circ$). When the material is oriented perpendicular to B_0 ($\theta = 90^\circ$), the effective component of the dipole moment along the field direction vanishes. This follows from the projection of the dipole moment vector onto the magnetic field axis.

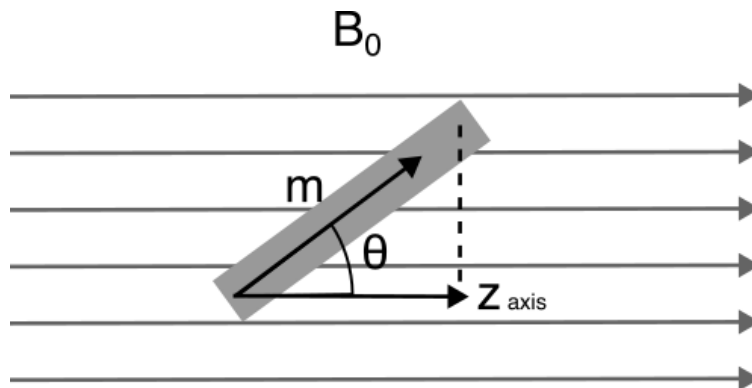


Figure 2.13: Vector diagram of a magnetic dipole $\mathbf{m}$ in a uniform magnetic field $\mathbf{B}$ directed along the z -axis.

In Figure 2.13, the dipole of length a , experiences equal and opposite forces on its “horizontal” sides, producing no net force but a torque magnitude $|\boldsymbol{\tau}| = |\mathbf{m} \times \mathbf{B}| =$

$mB_0 \sin \theta$. And a visual representation of the component of $\mathbf{m}$ along the field direction is given [30].

From the geometric projection of the dipole moment onto the magnetic field direction (eq. 2.16), and the torque experienced by the dipole we can write

$$|\boldsymbol{\tau}| = m_0 B_0 \sin(\theta) \cos(\theta) = m_0 B_0 \frac{\sin(2\theta)}{2},^1$$

using equation 2.10 for the dipole moment, we then write

$$\tau(\theta) = \tau_{max} \sin(2\theta), \quad (2.17a)$$

$$\boxed{\tau_{max} = \frac{\chi_m B_0^2 V}{2\mu_0}} \quad (2.17b)$$

A problem to overcome is that, as explained in section 2.1, epicardial leads are malleable (they can be bent or curved into any shape). The model above assumes a rigid dipole that spins uniformly with the magnetic field (B_0), for that reason a rigid torque transfer arm should be considered and its description is found on section 3.4.2.

The magnetically susceptible test piece experiences a magnetic torque following equation (2.17) and is exerted around the attachment point. The torque transmitted through the joint becomes a force, that can be measured with a sensor located at a known distance (r_{sensor}). The relationship of this force to the torque is

$$\tau_{max} = F_{sensor} r_{sensor} \quad (2.18)$$

since the sensor stops the rigid arm making an angle of 90 degrees and this behavior is later confirmed in this document (See Fig 4.7). Because torque is conserved along the rigid body, it provides a practical means to extend the moment arm beyond the

¹Using the trigonometric identity $\sin(2\theta) = 2 \sin(\theta) \cos(\theta)$.

base geometry ensuring sufficient space for accurate force sensing while maintaining constant torque transmission.

Furthermore, the sensor design will determine the signal resolution. In this case, a thin rod stops the arm and bends following the cantilever beam deflection equations. These equations indicate that the strain at the fixed end increases linearly with both the applied force and the beam length, suggesting a mechanical advantage for measuring small forces. Then the strain on the sensor caused by the force applied is measured via strain gauges leading to resistance change. A half-bridge Wheatstone circuit is configured the output voltage corresponding to strain on the cantilever (see Figure 2.16). The system is linearly calibrated by applying known weights to the free end of the cantilever, yielding the expected relationship

$$\Delta V = m\Delta W \quad (2.19)$$

where m , the slope of fit, is the calibration factor between voltage change (ΔV) into the corresponding change in applied force (ΔW Weight). Implementation of this sensor can be found on section 3.3.3 and a further investigation on how small of a force can these devices measure is given in section 2.2.6.

Once the torque is found, a direct comparison to the body's weight can be made. According to ASTM F2213 [23, 25], the criterion for evaluating magnetically induced torque on medical devices in a magnetic resonance environment is

$$c_\tau = \frac{\tau_{max}}{mgL}, \quad (2.20)$$

where c_τ is this criteria, τ_{max} is the maximum torque measured or obtained by equation 2.17b, meaning that the maximum magnetically induced torque is comparable than the product of the device's longest dimension (L) and its weight (mg). Under the

ASTM gravity-based acceptability criterion, a device is considered acceptable when $c_\tau < 1$.

2.2.5 Smallest measurable translational force

The smallest measurable translational force can be obtained with geometry. The pendulum forms a right triangle where the vertical line is the pole itself " L ", the horizontal is the ruler " x " and the hypotenuse is the pendulum's string as depicted on figure 2.14. Both of these measurements have a 1mm precision, yielding a $L=1\text{m}$ and $z=1\text{mm}$ triangle. Then, the smallest deflection angle capable of measurement is given by $\alpha = \tan(z/L) \sim z/L$ for small angle which is about of a 10^{-3} rad (~ 0.057 deg).

Since the angular deflection corresponds to the ratio of magnetic to gravitational forces (see eq. 2.12) the smallest measurable magnetic force is

$$F_{m,min} = \tan(\alpha_{min}) = mg\left(\frac{z}{L}\right). \quad (2.21)$$

For a 1g test mass², this corresponds to a detection threshold of $F_{m,min} \sim 10^{-6}\text{N}$ ($1\mu\text{N}$). Thus, the apparatus can resolve magnetic force as small as one-thousandth of the object's weight. This resolution defines the lower limit of measurable magnetic susceptibility using the pendulum method. For nonmagnetic or weakly paramagnetic materials, such sensitivity ensures that even subtle translational forces can be quantified, making the setup suitable for assessing device safety under MRI static field gradients.

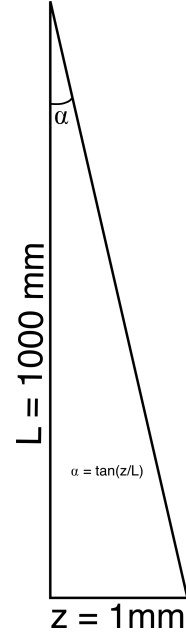


Figure 2.14: Right-angled triangle formed by the pendulum geometry at equilibrium.

²Sample calculations can be found on appendix B.2

2.2.6 Smallest measurable torque

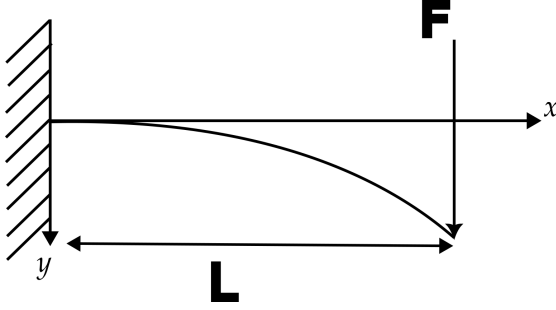


Figure 2.15: Cantilever Beam of length L with concentrated load F at the free end.

For torque measurements, the sensitivity of cantilever-based torque sensors is determined primarily by the relationship between the applied tip load and the strain induced at the fixed end of the beam. Figure 2.15 shows a diagram for a thin cylindrical beam of length L , diameter D , and Young's modulus E , fixed at one end and subjected to a transverse point load F at

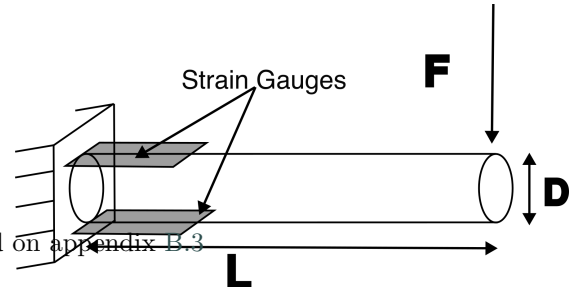
the free tip, classical beam theory [46] gives the maximum surface strain as

$$\epsilon = \frac{32FL}{\pi ED^3}. \quad (2.22)$$

This relationship indicates that the strain is inversely related to the Young's modulus and the cube of the diameter and directly proportional to beam length. As a result, longer and thinner rods increase the observable signal by producing greater stresses for the same applied force.

Following the geometry used by Chiu *et al.* [47], a wire of $L = 69.85 \text{ mm}$ and $D = 0.64 \text{ mm}$ fabricated from Nitinol ($E = 66.2 \text{ GPa}$) achieved an empirical detection limit of 0.49 mN , corresponding to an output change of approximately 5 mV . Using Equation 2.22³, this translates to a geometric sensitivity factor on the order of $L/D^3 \approx 2.66 \times 10^8 \text{ m}^{-2}$.

A similar configuration can, in principle, achieve micro-Newton sensitivity when paired with a low-noise amplifica-



³Sample calculations on this limit can be found on appendix B.3

38 Figure 2.16: Cantilever setup showing the fixed cylinder and strain gauges "sandwiching" the rod at its base

tion circuit, provided the strain gauge and readout maintain linear response and the system noise floor remains below the equivalent strain corresponding to the smallest measurable force.

A metallic foil strain gauge bonded to the beam surface, which its position is shown in Figure 2.16, converts the mechanical strain into a measurable change in electrical resistance. For small elastic deformations, the fractional change in resistance is linearly related to strain through the gauge factor GF [48, 49, 50]

$$\frac{\Delta R}{R} = GF \epsilon, \quad (2.23)$$

where GF is typically in the range of 2.0–2.2 for constant foil gauges. Combining Equations 2.22 and 2.23 gives the total resistance change due to an applied tip force.

$$\Delta R = R_0 GF \frac{32FL}{\pi ED^3}. \quad (2.24)$$

This resistance change is converted to a differential voltage when the gauge is part of a Wheatstone bridge circuit. For a half-bridge configuration, utilized by Chiu, *et al.* [47], where two active strain gauges form adjacent arms of the bridge, the small-signal output voltage under excitation V_{exc} is approximately [48, 51]

$$V_{out} \approx \frac{V_{exc}}{2} GF \epsilon, \quad (2.25)$$

Assuming perfect linearity, little temperature drift, and perfect bridge balancing. The direct proportionality between applied force and bridge voltage may be obtained by substituting Equation 2.22:

$$V_{out} = \frac{16 GF V_{exc}}{\pi ED^3} L F. \quad (2.26)$$

In practice, the smallest detectable force is governed by the combined noise of the gauge, bridge, and amplifier system. If the noise-equivalent output voltage V_{noise} corresponds to a minimum detectable strain $\epsilon_{min} = 2V_{noise}/(V_{exc}GF)$, then the force detection limit follows from Equation 2.22 as

$$F_{min} = \frac{\pi ED^3}{32L} \epsilon_{min}. \quad (2.27)$$

Theoretically, the sensor can resolve forces in the micro-Newton range by optimizing the ratio L/D^3 , minimizing circuit noise, and utilizing materials with moderate modulus (such as aluminum or Nitinol). Then, assuming a bridge noise ($\sigma V = 0.0005$ V) and the calibration slope for a particular sensor the smallest torque ($\sigma\tau$) or limits of detection can be found. These restrictions can be decreased by averaging (σ scales as $1/\sqrt{N}$), by increasing mechanical leverage (slope grows almost linearly with cantilever length), or by enhancing the bridge/amplifier noise performance. For quasi-static measurements, the limits of detection may be moved from tens of mg into the single-mg or sub-mg range using a combination technique (longer/thinner cantilever + averaging + low-noise amplifier); dynamic measurements will be constrained by bandwidth and $1/f$ drift. In our lab we are limited by the materials available, meaning a test on different sensor was done.

2.3 Concluding Remarks

This chapter has reviewed the essential theoretical foundations for evaluating MRI safety in patients with retained epicardial leads. Epicardial leads, integral to CIEDs, differ from transvenous leads in placement, design (unipolar vs. bipolar, coaxial vs. coradial), fixation (suturing vs. screwing), and material composition (paramagnetic electrodes and conductors; diamagnetic insulation). Their direct attachment to the epicardium and frequent abandonment (often as 5–30 cm fragments, 0.5–1.5 mm di-

ameter) heighten concern for mechanical interactions with MRI fields, despite low-to-moderate magnetic susceptibilities that suggest minimal risk.

MRI fundamentals, static field (B_0), RF excitation, gradient encoding, and relaxation (T_1 , T_2), were explained aiming to provide enough information to assess the interaction with implants. The emphasis was on the static field, capable of inducing translational forces and torques on conductive implants, and the physics that rule the interaction seeking to characterize such forces.

The relevant expressions used throughout this work are summarized below for clarity

Summary of Relevant Expressions	
$F_m = \frac{\chi_m V}{\mu_0} B_0 \frac{dB_0}{dz},$	(Eq. 2.11)
$\frac{F_m}{F_g} = \tan(\alpha),$	(Eq. 2.12)
$\tan(\alpha) = \frac{\chi_m}{\rho \mu_0 g} B_0 \frac{dB_0}{dz},$	(Eq. 2.14)
$\tau(\theta) = \tau_{max} \sin(2\theta),$	(Eq. 2.17a)
$\tau_{max} = \frac{\chi_m B_0^2 V}{2\mu_0},$	(Eq. 2.17b)
$\Delta V = m \Delta W,$	(Eq. 2.19)

Figure 2.17: Summary of relevant equations used in this study.

Current guidelines (HRS 2017, ESC 2021, CMS) remain conservative, considering systems with abandoned or epicardial leads non-MRI-conditional due to insufficient data, despite evidence that post-2000 leads may be safe under monitoring [16, 18]. This gap motivates the present experimental study. By quantifying static-field forces and torques in controlled settings, using ASTM-based pendulum and torque-transfer methods (resolving forces and torque to $\sim 10^{-6}$ N), the work aims to challenge restrictive policies and enable evidence-based updates to clinical MRI protocols for this growing patient population.

Chapter 3

Materials and Methods

This chapter details the experimental setups and measurement procedures developed to quantify MRI-induced translational forces and torque acting on post-surgical epicardial pacing lead when exposed to its static magnetic field. Three non-magnetic test apparatus were constructed. Specifically designed to measure these forces with the precision required for clinical applications, in accordance to ASTM guidelines.

Given the challenges associated with clinical access for experimental testing, as MRI scanner time is typically associated with long scheduling delays and short available measurement periods, a secondary objective of this work was to improve the quality of our experimental setup in order to maximize the efficiency of data acquisition during these limited observation windows. This chapter describes the experiments that were conducted and the methodologies implemented.

Previous students of Creighton University have developed the laboratory-scaled instruments and associated software necessary to evaluate magnetic forces in a controlled environment. This was achieved by generating a smaller but reproducible magnetic field in the laboratory, enabling characterization of ferromagnetic materials. While this laboratory-generated field was not sufficient to directly measure or characterize the epicardial lead itself, it allowed for evaluation of the accuracy and

measurement capacity of each instrument, as well as for the observation of the physical phenomena described in sections 2.2.3 and 2.2.4 with its relevant expressions in equation 2.3. Importantly, both translational force and torque scale predictably with the magnetic field strength, specifically, $F \propto B_0 \frac{dB_0}{dz}$ and $\tau \propto B_0^2$, which allows the laboratory setup to reproduce the same dimensionless relationships observed under clinical (3T) conditions, merely at smaller magnitudes. Therefore, by appropriately scaling χ_m and ρ (see Appendix B.1), the essential magnetic behavior can be simulated without requiring a full-strength MRI system.

Three distinct experimental setups were involved in this study: (1) a spatial field mapping platform to characterize B_0 and ∇B_0 , (2) a pendulum-based deflection apparatus to assess translational force through angular displacement, and (3) a torque measurement system for rotational force assessments. In this chapter we will explain our setup and rationale, as well as the proven modifications to the instruments in order to provide a reproducible map of action to test our instruments.

3.1 Magnetic Field Generation and mapping

Seeking to archive the most reliable or similar situation as one would encounter on a clinical MRI, magnetic fields in the lab were generated using two approaches. The most intuitive methods involved either neodymium permanent magnets or a copper-wound solenoid. Each approach presents advantages and limitations. For example, a neodymium magnet produces a strong, fixed at its maximum value, magnetic field (B_0) with distance being the only parameter that affects its strength. Much like an actual MRI). In contrast, the magnetic field of a solenoid is dependent on the current flowing through its wire, allowing for fine changes of magnetic field (B_0) without physically relocating the magnet or the probe.

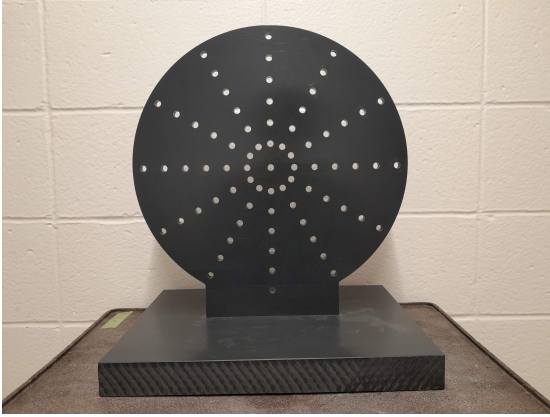
The approach used in this section was the later. A custom-built air-core solenoid of approximately 15 cm in length and 10 cm in outer diameter was densely wound with copper wire (~ 1 mm diameter) meaning a winding density of 50,000 turns per meter, forming an inner core of approximately 5 cm. A Pasco Scientific SF-9584 DC power supply was used to deliver currents in the range of 0–4 A, corresponding to magnetic field strengths from 1 to 18.9 Gauss at about 8cm for this solenoid.

Magnetic field strength (B_0) and gradient (∇B_0) measurements were performed using a commercial Gaussmeter (Model GM2, AlphaLab Inc., Salt Lake City, UT). The GM2 operates on the Hall effect principle, wherein a voltage is generated across a probe when exposed to a perpendicular magnetic field. This voltage is proportional to the local magnetic flux density, enabling precise spot measurements of field magnitude. The Gaussmeter used has an accuracy of 1% of the DC reading in the 16°C to 29°C range.

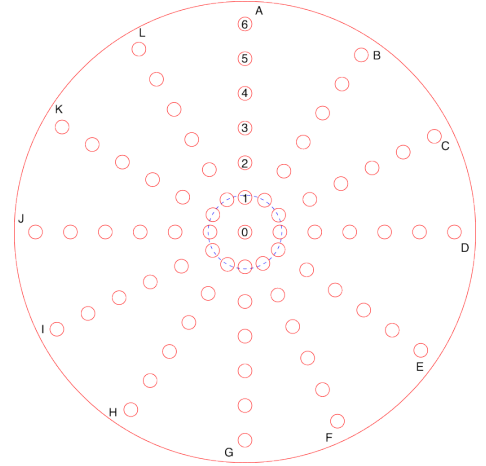
In the lab, our solenoid center sits at 11 cm from a optical breadboard. At its front, three positions were selected to place the gaussmeter probe to check for homogeneity, at the center ($r=0$ cm, $\theta=0$), southwest ($r=10$ cm, $\theta=225$), and east ($r=10$ cm, $\theta=0$). This was achieved by using a translational station positioning the probe 10 away from the solenoid’s entrance. The station uses 3 servomotors, one for each Cartesian axis, to accurately position the probe with changes less than a tenth of a millimeter. However, the distance across the z-axis was limited to 11 mm by the servo’s maximum extension. This setup was used for experiment in section 3.2. To make a quick check across a wider range, a ruler was fixed at the same three positions described and the probe was steadily translated about 20 cm to the solenoid entrance and back. Data was recorded using the manufacturer’s default software, AlphaApp with a sampling time of 1 reading per second, for about 2 min. Video recording allowed to map time, and distance enabling a rough map of the B field and its gradient.

3.1.1 Magnetic Field and Gradient Mapping Platform

To ensure spatial accuracy and reproducible measurements, a fixture made from rigid PVC was constructed by previous Creighton University students [29]. The platform consists of a $300(\text{W}) \times 420(\text{L}) \times 40(\text{H})$ mm rectangular prism as base and a 280 mm diameter vertical circular plate, perforated with a polar grid of probe-access holes, with 12 radial lines spaced at 30° . Each line contains 5 holes at uniform radial spacing of 20mm, sharing a single central hole as shown in Figure 3.1 for 49 different positions. Each access point was numerically indexed to facilitate 2D field mapping in the X-Y plane. This fixture has a total weight of about 9kg and height of 35 mm positioning the center at 21 mm to align with the MRI’s isocenter.



(a) PVC-based mapping platform.



(b) Example field map output from MRI fringe region.

Figure 3.1: PVC-based magnetic field mapping platform. Used for spatial B_0 and ∇B_0 measurement. based on design from Ferreira, et al. (2017) [29, 21].

Once the field-mapping fixture and Gaussmeter probe are co-aligned, data capture is streamlined using LabVIEW to be sufficiently flexible and precise for our proximity measurements near an MRI. The manufacturer’s default software, AlphaApp, provides basic data-logging, recording, live screen streaming, and plotting of measurements from USB-enabled AlphaLab meters[52]. However, AlphaApp’s control over sampling frequency is limited by design, therefore limiting fine temporal resolu-

tion when the probe is moved into steep field gradients. In contrast, using LabVIEW Virtual Instrument (VI) interface allows for configurable sampling intervals, buffer control, and precise timing protocols, enabling us to push the device for much higher accuracy as the probe approaches and crosses isocontour regions within the MRI fringe. We developed that VI by leveraging the AlphaLab Data Acquisition Communication Protocol [53] to command the device directly via USB series port, providing control over timing and sample rate. The data acquisition procedure for magnetic flux mapping is organized in this VI, whose basic logic is summarized in Appendix .

In clinical use, the entire fixture is intended to be shifted axially along the Z-direction of a horizontal bore MRI scanner using controlled couch increments (e.g., 10 cm). This translation allows for volumetric sampling of B_0 over several planes. The spatial magnetic field gradient ∇B_0 can then be estimated numerically using finite difference calculations between adjacent measurement points, as developed in Chapter 2.2.3. This method has previously been applied to MRI mapping in the work by Ferreira [21].

3.2 Translational Force Measurement via Angular Deflection

The solenoid magnet was particularly useful for generating a variable magnetic field for the translational forces measurement tests. It allows for a range of magnetic fields strengths (B_0) at a given distance from the magnet without the need of displacing our fixture. The magnetic field was measured at a stable point along the pendulum trajectory for a given current or magnetic field (B_0) and then half a millimeter around that point to find the gradient as per eq. 2.8. The gaussmeter in the VI was operated in a single mode; with rest periods between trials due to the

time required to stabilize the pendulum after each adjustment. This methodology is detailed below.

3.2.1 Measurement Device and Test subjects

To assess MRI-induced translational forces in a laboratory environment, our lab developed a pendulum-style apparatus following the geometry and physics outlined in ASTM F2052-15 [5], showcased in Figure 3.3. The fixture itself predated this project, but no engineering drawings or dimensional records existed. For reproduction and future reconstruction, a representation of it was reconstructed in CAD so that future students or labs can rebuild the geometry without much guesswork. These reconstructed views are shown in Fig. 3.2 with design engineering drawings in appendix A.1.

The experimental frame was constructed entirely from PVC components, including a vertical post of about 1.5 meters and a rigid base of dimensions $30(L) \times 20(W) \times 4.5(H)$ cm for stabilization. The vertical post includes a set of two rings fixed with nylon screws, to adjust heights, a default suspension height of $L = 1$ meter between these two rings was used for consistency across trials, maintaining the geometry and proportions showed in equation (2.21). This distances was measured from the last contact point of the string to the rubber on the top ring to the sample hanging free at ruler's height. The lower ring, in Figure 3.2(c), features a slit where a custom made Kydex™ ruler sits held by smaller nylon screws. This ruler is interchangeable. The upper ring in the CATIA model (Fig. 3.2(b)) was simplified for demonstration, however the final implementation shows a rubber clamp from which a string holds.

For clarity, Fig. 3.3 shows the corresponding photographs of the actual apparatus, with close-ups of the same features documented in the CAD reconstruction.

Testing for low field lab measurements a solid ferromagnetic cylinder was used as sample, showed in figure 3.4. Its dimensions were captured with caliper, ruler, and

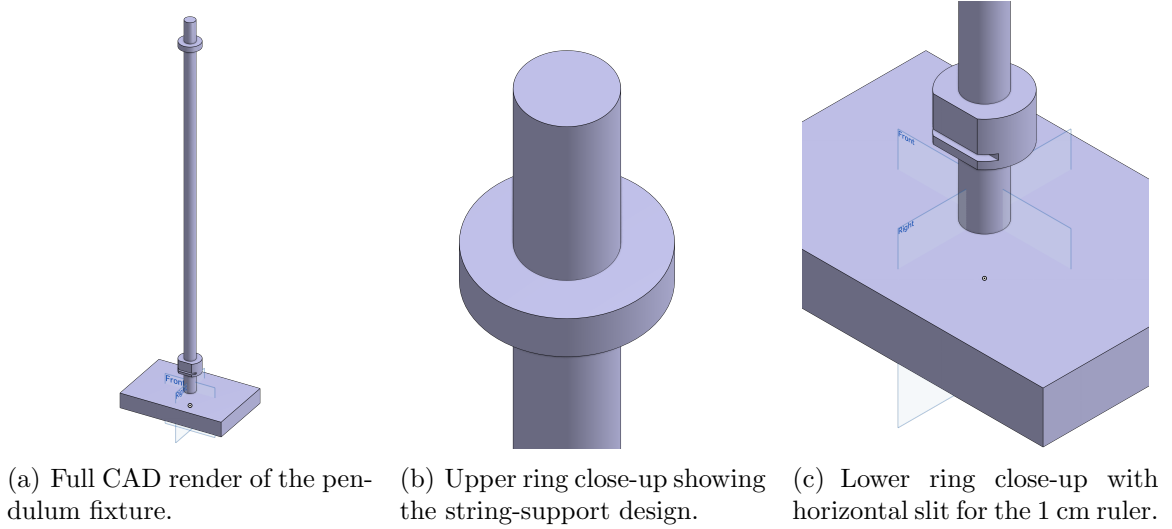


Figure 3.2: CAD render and detailed component views of the pendulum fixture used for translational force testing. Panel (a) shows the full assembly. Panel (b) highlights the upper ring designed to secure the pendulum string. Panel (c) shows the lower ring containing the slit used to align a 1cm-wide ruler for displacement measurements.

scale, yielding length = 6 mm, diameter = 3 mm, and mass = 0.3648 g therefore density $\rho = 8.6 \text{ g/cm}^3$. The sample was suspended by a 955 mm-long sewing thread, tied with a simple knot at its center. For clinical applications, a 0.25 mm nylon monofilament is recommended. A monofilament nylon thread has low stretch, complies with MR-safe properties (Non-metallic, Non-conductive, Non-magnetic), and reduces variability introduced by torsional drift or thread elasticity [25, 5]. The bottom of the sample was free to swing, forming a pendulum under gravity effectively aligning with model on figure 2.12 and equation 2.11. A horizontal ruler, fixed along the Z-axis, served as the visual reference to quantify lateral displacements. The fixture final form looks like this

To accommodate epicardial lead positioning while maintaining MRI compatibility, a lightweight, non-magnetic bee comb mesh was developed and fabricated from Polylactic Acid (PLA). The mesh dimensions were 16 (L) $\times$ 14 (W) $\times$ 1 (H) mm, for a total volume of less than 100 mm^3 equivalent to around 100 mg given the 1.24 grams per cubic centimeter (g/cm^3) PLA density. Figure 3.5 shows a 3D render of



Figure 3.3: Full and detailed views of the pendulum fixture.

the mesh, with the desired proportions. This mesh was 3D printed and has a final weight of 130mg. The mesh was suspended such that the lead axis was aligned approximately parallel to the magnetic field gradient as showed in Figure 2.8. The mesh’s lightweight design aimed to minimize its contribution to the total magnetic force, however its mass was not explicitly subtracted from the translational force calculations.

A summary schematic of the apparatus and frame are included in Figures 3.2 and 3.5, with additional engineering drawings available in Appendix A.

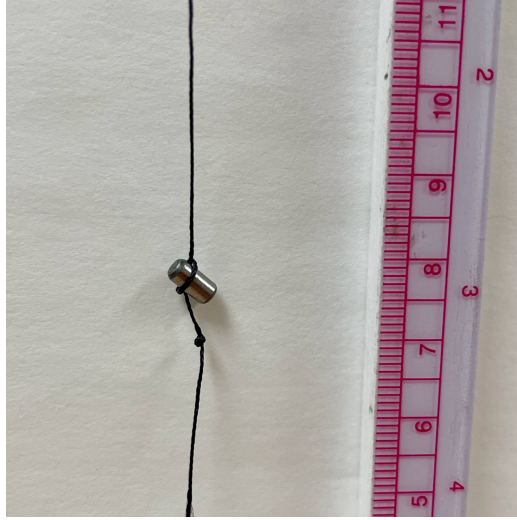


Figure 3.4: Ferromagnetic Subject and string attached, a ruler in centimeters and inches is showed for size reference

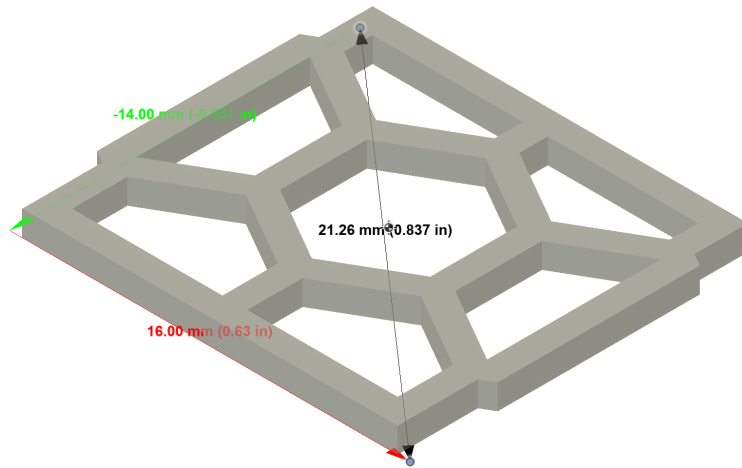


Figure 3.5: The design on lead holder was made as a rectangular lightweight bee comb mesh to wrap the epicardial lead around it. It holds in a pendulum with a string

3.2.2 In Lab Measurement Protocol

Magnetic field measurements were performed using the AlphaLab Inc. Gaussmeter described in section 3.1. The setup used our custom-made solenoid connected to a Pasco power supply capable to deliver up 8 A acting as variable magnetic field generator for our experiment. The subject, as described in section 3.2.1, was attached to a string pending from the MRI safe pole and angle of deflection was measured

visually with a ruler on the subject horizontal displacement towards the coil. The Gaussmeter was positioned such that it allowed for small displacements and magnetic field measurements at the same time as angle deflection.



Figure 3.6: In-lab experimental setup for MRI translational force assessment.

Figure 3.6 shows the experimental lab setup used to assess the methodology accuracy and reliability with all used components. On the left side sit the magnetic field generation instruments, whereas on the right side a gaussmeter with the probe on its translational stage allows for sub-millimeter displacement along the z-axis. As briefly explained in section 3.1 the translational stage uses three Thorlabs travel motorized actuators driven via the Kinesis software interface. While Kinesis-controlled stages support sub-micrometer incremental motion and repeatability on the order of microns[54], our experiment only required positional accuracy on the order of 0.5

mm. Thus the stage system offered more than sufficient precision and stability for repeatable setup between trials. Inside Kinesis the speed was set up for 0.3 m/s and the maximum displacement of each actuators was 12 mm. At the center sits the main device ready for clinic use, in Figure 3.7 an example of the working mechanism is depicted. Initially, the subject remains at rest (Figure 3.7(a)) once the current is flowing the magnetic field on the solenoid pulls the subject towards it (Figure 3.7(b)).



(a) Subject hanging in pendulum at rest.



(b) Translational force pulls the subject towards the magnetic field source.

Figure 3.7: Comparison of the subject behavior with and without translational forces during magnetic field exposure.

Initial trials attempted to map the magnetic field from the furthest position of the pendulum, 8 cm from the solenoid face, at a constant solenoid current in a single run. The idea was to obtain (B_0) at the pendulum's resting position and then compute the field gradient at any point from that map. Afterward, the pendulum was allowed to be slowly pulled toward the solenoid to its equilibrium position so its displacement (in cm) could be recorded. This was repeated while increasing the solenoid current in 0.5 A steps, aiming to fit the resulting equilibrium positions using Eq. 2.14. These attempts produced inconsistent results that pointed to errors inherent to the setup. Because the current, rather than the entire apparatus, was changed, the value of B_0 at the pendulum's resting position did not correspond to the required by Eq. 2.14. Even using the mapped field, estimating the correct value for the magnetic field for

each equilibrium position was unreliable. Moreover, predicting equilibrium positions before running the experiment proved difficult. Additional complications came from solenoid imperfections, current leakage, and induced currents, all of which became significant once displacements exceeded 1 cm. At higher fields the pendulum was sometimes pulled fully into the solenoid. Another factor was the gaussmeter restriction on accurate motion tracking to roughly 11–12 mm on the translation stage. While this approach resembles the proposed clinical protocol, the scale of the apparatus and the nonuniform fields make it impractical. Clinically, a single mapping session would give both B and ∇B at any point, leaving the angle measurement as the critical step.

The in-lab methodology that ultimately worked inverted the logic of the trials. Instead of fixing the current and observing displacement, the displacement itself was fixed. We chose increments of 1 mm from the resting position up to 11 mm. This solved several issues simultaneously. Induced currents remained low because the ferromagnetic body barely moved, the translation stage’s full range was used, the subject never entered the solenoid unintentionally, and the subject’s position was always precisely known. For each displacement, the current required to pull the pendulum to that point was measured and recorded. A magnetic-field map was then acquired at each corresponding current level, ensuring the equilibrium position and its ± 0.5 mm neighbors were carefully sampled. With this method, the value of B_0 at equilibrium was known for every current level. A faster variant of this procedure would measure only at the equilibrium position and within ± 1 mm around it. Appendix E includes sample data from these trials.

The measurements taken for characterizing the local field gradient, at the pendulum’s equilibrium position (B_0), 0.5 mm below ($B_{-0.5}$), and 0.5 mm above ($B_{+0.5}$) were sampled at least nine times per current level, and gradients were computed using Python pandas library with the central finite difference method (Eq. 2.15) producing

$$\frac{dB}{dz} \approx \frac{B_{+0.5} - B_{-0.5}}{1 \text{ mm}}, \quad (3.1)$$

The 1mm increments in displacement was translated to angular displacement as

$$\alpha = \tan^{-1} \left(\frac{x}{L} \right) \quad (3.2)$$

Allowing for plotting and interpretation with equation 2.14.

All measurements were logged in Microsoft Excel, and photos/videos were captured via smartphone to document behavior. A minimum of three trials were performed for each current level; however, only the final (most consistent) dataset was retained for analysis. Data were exported to a Jupyter Notebook for clean up and processing with pandas. Then plotted to evaluate $\tan \alpha$ vs. field gradient force. Propagated uncertainties in α and dB_0/dz were computed using standard error propagation formulas.

3.2.3 In Clinic Measurement Protocol

To guarantee safety and repeatability, the measurement of MRI-induced translational forces on the test items was carried out in accordance with a structured methodology. Initially, all personal belongings were removed to prevent safety issues. The apparatus, a PVC pole equipped with a non-magnetic ruler aligned with the pendulum string, was positioned at a precise distance from the MRI bore. Two sets of measurements were performed sequentially: first using a magnetically susceptible test object, and then with the epicardial lead attached, suspended from a light basket and a heavy basket, respectively, to assess force variations. Moving closer to the bore, a step-wise method was followed to measure displacement at certain translational points. Each measurement phase was completed as per the flowchart-guided

process, with detailed data recorded in the a worksheet. The comprehensive protocol, including step-by-step instructions and safety considerations, is provided in Appendix

3.3 Sensor and Circuit Design for Torque Measurements

A sensor paired with an amplifier circuit made by previous student followed the ideas described in Chiu *et al.*(2016)[47] work. Distinctively, the introduction of the ability to zero the reference level at 0 V with two trimming potentiometers was implemented. This work predates this document and exists only as lab-notebook entries and circuit sketches. Relevant portions are included both as documentation and as the starting point for the retesting and characterization carried out.

3.3.1 Sensor Design

Figure 3.8 shows a sample from a set of seven beam force sensors were produced with in-lab materials, the sensor is conformed by two main pieces the beam itself and a pair of OMEGA[®] precision strain gages (SGD-3/350-LY11) [55], as reviewed in section 2.2.6. This strain gages were glued on opposite sides of the rod around 3 to 5 cm from its end. This two variable resistances (strain gages) are part of a Half-Wheatstone bridge configuration and are connected to the circuit with soldered jumper wires, see section 3.3.2. The beam materials included glass, aluminum, titanium and two types of plastic. Table 3.1 summarizes their characteristics, they were made of similar sizes to test repeatability. Two of the seven available sensors were found to be irreparably damaged, so no further measurements were possible on them.

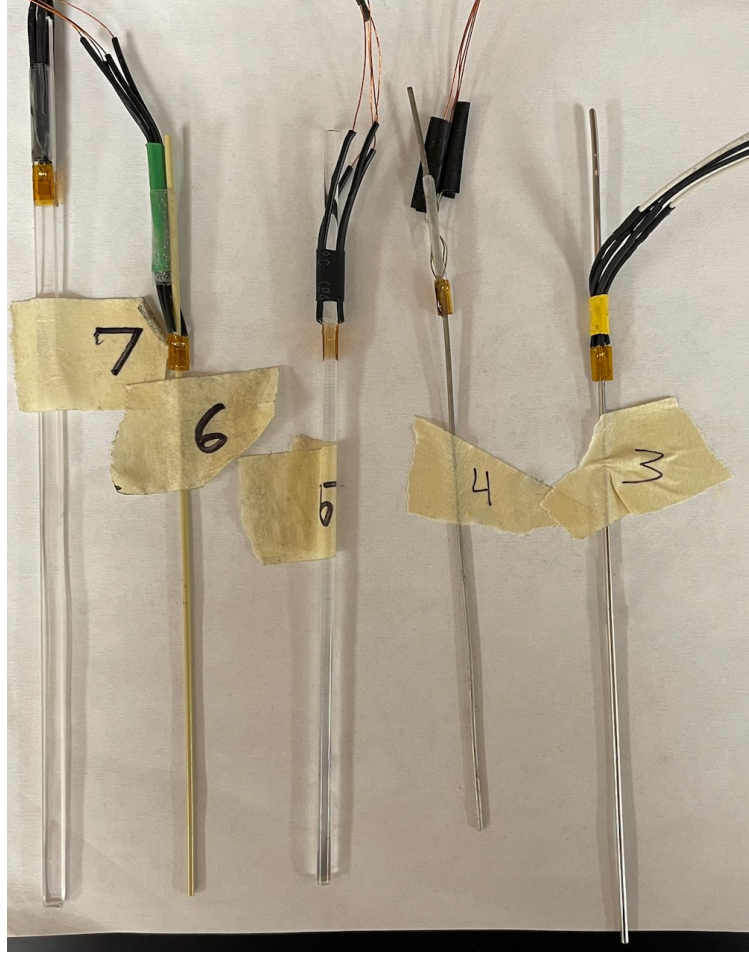


Figure 3.8: Rods fabricated during the torque experiment design. From left to right, (7) Clear Plastic, (6) Yellow plastic², (5) Glass, (4) Titanium ¹, (3) Aluminum

Table 3.1: Characteristics of the rods used in experiments.

Rod ID	Material	Diameter (mm)	Length (mm)	Remarks
Rod1	Aluminum (Al)	1.6	145.0	
Rod2	Al	1.6	182.00	
Rod3	Al	1.6	182.00	
Rod4	Titanium	< 1.6	220.00	Not practical ¹
Rod5	glass	3.2	166.0	
Rod6	yellow plastic	1.6	220.00	Visible damage ²
Rod7	clear plastic	2.9	180.0	

¹This rod was originally stored on a spool and had to be straightened manually. That process introduced micro-cracks, so the material no longer behaved elastically and deviated from Hooke's law.

²This rod was discovered broken in the circuitry and was noticeably bent, indicating the need for full refabrication.

3.3.2 Circuit Design

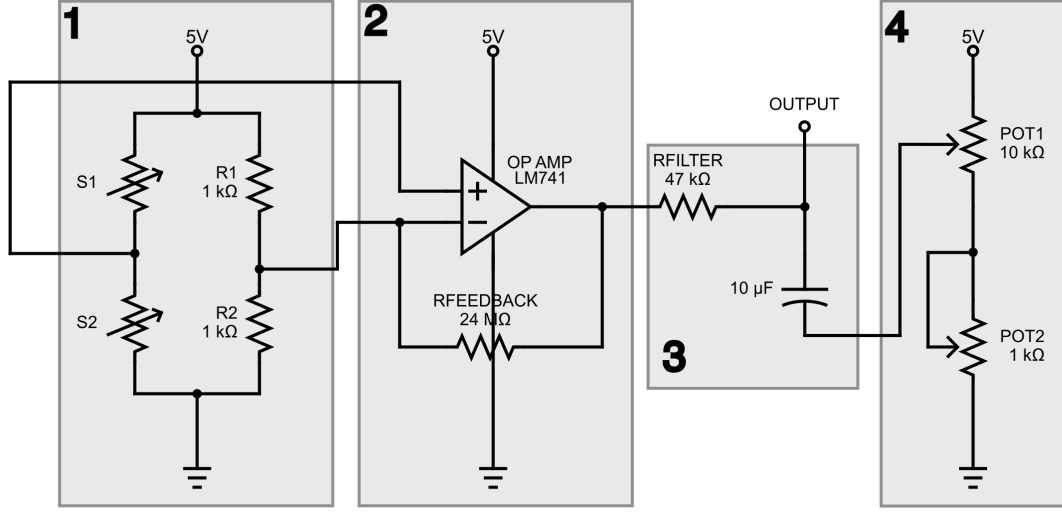


Figure 3.9: Circuit developed for the torque measurement setup, following the design principles of Chiu *et al.* (2016)[47]. Features include, 1. half Wheatstone Bridge Strain Gauge, 2. Amplifier, 3. Low-pass filter, 4. Coarse-fine adjustable reference network.

The measurements for the experiment in section 3.4 were done utilizing the already existing circuit prototype, a description and electronic analysis is given that followed its scrutiny. Figure 3.9 shows the circuit main parts, it includes four main sections, the Wheatstone bridge, the amplifier, a low pass filter and the ground offset potentiometers. From a Kirchhoff analysis for the first two sections, the transmission function was derived as

$$\frac{V_{out}}{V_{in}} = \frac{R_f}{R} \frac{S2 - S1}{S2 + S1} + \frac{S2}{S2 + S1} \quad (3.3)$$

where $R = R1 = R2$, $S1$ and $S2$ are the strain gauge and R_f is the feedback in figure 3.9. The output voltage of the circuit is driven by the relative difference between the two strain gauges compared to their total resistance. The operational amplifier multiplies this small imbalance by the gain factor $\frac{R_f}{R}$. The bridge sets a baseline through the voltage division $\frac{S2}{S2+S1}$, whereas the amplified differential term $\frac{S2-S1}{S2+S1}$

determines the sensitivity of the circuit to strain. For the complete derivation see appendix B.4.

After the instrumentation–amplifier stage, the output passes through a first–order RC low–pass filter composed of a 47 k Ω resistor in series with a 10 μ F capacitor to ground, as in Figure 3.9 part 3. The time constant is then $\tau = 0.47$ s meaning a 0.33 Hz cutoff frequency. Finally, a first–order system with this time constant has a 10 $\rightarrow$ 90% rise time of approximately 1.03 s. In other words, whatever the Wheatstone–op–amp transfer function generates, this RC stage only slows its dynamics rather than altering its steady–state gain. It suppresses high–frequency noise from the strain gauge bridge and smooths fast fluctuations that the rod could not physically produce anyway.

In part 4 of Figure 3.9, the DC reference entering the filter is set by the two potentiometers analyzed in Appendix B.5. Both pots sit in series, forming a programmable voltage divider from 0 to 5V, and the capacitor of RC network simply charges toward whichever reference these pots define. From the voltage-divider derivation in Appendix B.5, the node voltage obeys

$$V_{\text{ref}} = V_{\text{in}} \frac{R_2}{R_1 + R_2}, \quad (3.4)$$

so sweeping the fine pot moves R_2 over a narrow range (~ 0 –1 k Ω), which shifts the reference only within a small, controllable window. The coarse pot changes the large R_1 segment (~ 10 k Ω), expanding or collapsing the overall span.

3.3.3 Sensor Calibration

Prior to any torque measurement, each sensor or rod from table 3.1 was tested or calibrated to accomplish two things, ensure use of the most sensible sensor and to acquire the conversion factor or slope of voltage to weight explained in section 2.2.5.

Additionally orientation and position of the rods was tested to ensure optimal values on these parameters for torque measurements.

For calibration in the lab, sensor was set in a cantilever position as shown in Figure 3.10, so the bar face horizontally and a small cone-shaped basket made out of tape was attached to hold increments of water. Water was added in μL increments (equivalent to milligrams) simulating torque loads from 0 to 800 mg. Initial increments of 10 up to 200 μL covered the expected force range of epicardial leads weight, followed by 50 μL increments to demonstrate linearity and sensor stability.

The change in voltage and water is then plotted and a fit can be found, see Section 4.3.1.

Then, in order to find the best position for sensor we set the following experiment, a set of neodymium magnets were aligned to the ceramic bearing to get a static magnetic field, and a platform angled at 45° carried the sensor as we moved it inwards. From section 2.2.4, it was shown that the maximum torque appears at this angle. Then we tested for the optimal length (L) and position from the center (r) as shown in Figure 3.12(a). This process was repeated to find the sweet spot on distance from the center up to a 100 mm radius and the best height was tested to different lengths along the rod sensor.

3.4 Torque Measurement via Counter-torque

For the torque estimation, ASTM 2213-17 [23] lists five different methodologies. Calculation based on measured displacement force method, low friction surface method, suspension method, torsional spring method, and pulley method. In section 1.1, the results on three of these methods were revised. Those included low friction, pulley and a variation of torsional spring methods [22, 25, 24], which showed good results but further considerations are needed. In our case a torque measurement device,

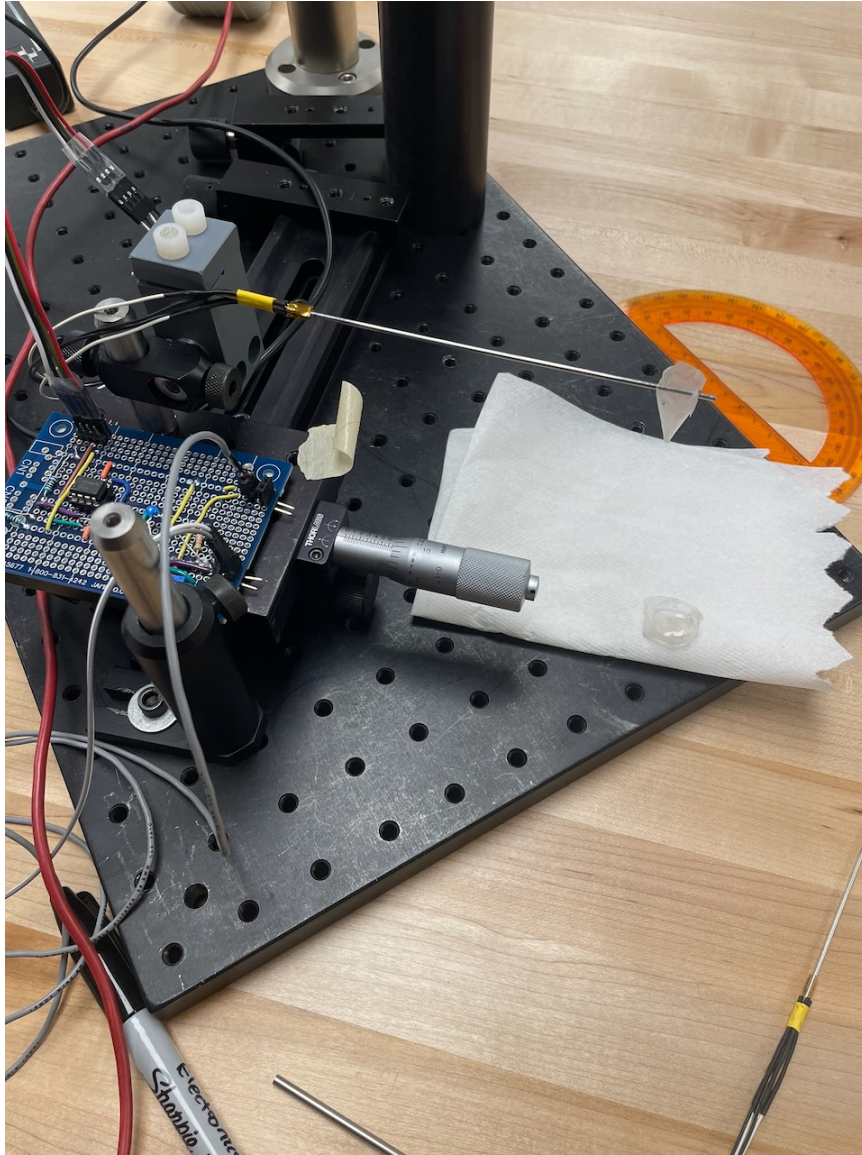


Figure 3.10: Circuit and sensor. The sensor was clipped with a custom made PVC clamp horizontally to act as a cantilever beam, a boat at length L was added, then a pipette was used to add water and see voltage changes with weight.

previously shown in Figure 3.12(a), capable of holding both the sensor and subject was developed and will be further explained in section 3.4.2. In addition to the sensor and circuit developed in section 3.3, a DAQ system and a computer with a VI was needed.

3.4.1 MRI Magnetic field conditions in lab recreation

For the initial configuration, a permanent-magnet–based torque setup was implemented as shown in Figure 3.11(a). This first arrangement employed four neodymium magnets, each magnetized through their thickness, with nominal dimensions of $1/4$ in thickness and $5/16$ in outer diameter. The magnets were mounted symmetrically on a rigid mechanical frame, with their magnetization axes oriented to produce a static magnetic field across the region occupied by the torque arm. Threaded lateral adjusters and a central support allow controlled positioning and alignment of the magnets relative to the rotating element.



(a) Initial permanent-magnet torque measurement setup. The apparatus employs four neodymium magnets symmetrically mounted on a rigid frame to generate a static magnetic field across the torque arm.



(b) Helmholtz coil configuration used to generate a controllable and approximately uniform static magnetic field at the coil center. The coils are driven with equal currents in the same direction to superpose their magnetic fields.

Figure 3.11: Comparison of the two magnetic field generation setups used for torque measurements: (a) a permanent-magnet–based configuration providing a fixed static field, and (b) a Helmholtz coil configuration enabling controlled variation of magnetic field strength.

A second setup (Figure 3.11(b)) utilized Helmholtz Coils EM-6722 from PASCO Scientific to create a controlled, nearly uniform static magnetic field that could be varied in strength and observed in real time. These Helmholtz coils consist of two identical coils, each wound with 200 turns, they were wired so current flowed in the same direction. They had an inner radius of about 10.06 cm and outer radius around 10.58 cm, and they sit on a optical base that lets us adjust the center-to-center spacing

close to where the field is most uniform (about one diameter apart). The coils used a maximum DC current of 2 A, which we supply with the PASCO Scientific SF-9584 DC power supply to increase the magnetic field incrementally; as current increases, so too does the magnetic flux density between the coils, allowing us to observe corresponding torque changes on our contraption.

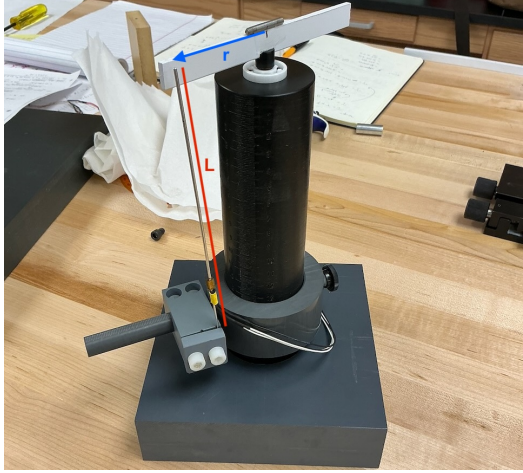
3.4.2 Torque Measurement Apparatus

On the final part of the assessment, the magnetically induced torque on a horizontally oriented subject was measured. To achieve this, a non-magnetic pole was fabricated with clinical considerations in mind. Figure 3.12(a) shows the pole, which was made from Delrin[™] and is 200 mm tall with 51 mm of diameter it can be attached to a PVC base for support with a nylon screw and it features a space for a non-magnetic bearing to sit. This non-magnetic ceramic bearing is meant to reduce as much friction for the rotation as possible while keeping everything non magnetic. Figure 3.12(b) shows a close up to the bearing, which has a plastic notch capable of holding a 10 mm thick ruler (lever arm) to spin.

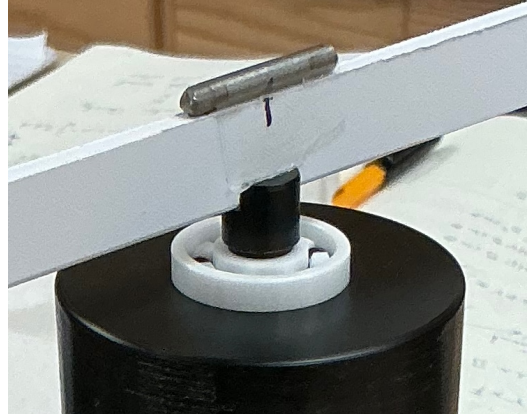
The induced torque on the subject, explained in section 2.2.4, is transmitted mechanically to a non-magnetic ruler which then would apply a force on the sensor introduced in section 3.3. This sensor would bend on and the counter-force would be measured using strain gauges.

To show the effects of the magnetic field on the setup, a big ferromagnetic dowel was positioned on top of the ceramic bearing. It was around 20cm long and with it a strong dipole moment could be induced, serving as a visual confirmation that the system is working.

However a smaller dowel is needed, this dowel would experiment much smaller torques based on size only. And following the same logic as in translational this



(a) This shows the pole and the lever arm of the torque fixture as it would be taken to the clinic, it labels 2 parameters (L red) and (r blue), which were adjusted for maximum torque measurements.



(b) A zoom-in to Figure 3.12(a), so the plastic notch on top of the bearing can be seen. it is holding the lever arm and the subject allowing for friction free rotation .

Figure 3.12: Main Torque fixture, this fixture was modified designed to include a base and a ring to hold the sensor. Previous work only included the main pole.

would be a better test subject at lab scale. With a small magnetic field, high magnetic susceptibility and small volume, following equation 2.19 proportionality is ensured.

In Lab Measurement Protocol

During the experiment, 2 measurements were performed. Magnetic field flux in Gauss using the AlphaLab Inc. Gaussmeter, described before, and Voltage from the strain gauge plus circuit sensor.

The experiment consisted of, with the sensor fixed in the ideal position, vary the magnetic field produced by the Helmholtz coils -mimicking the behavior in distance as a real MRI-increasing as we get closer to isocenter. Then the respective voltage produced by the deformation of the sensor is recorded. Finally, the voltage to weight calibration serves to estimate the induced torque of the system.

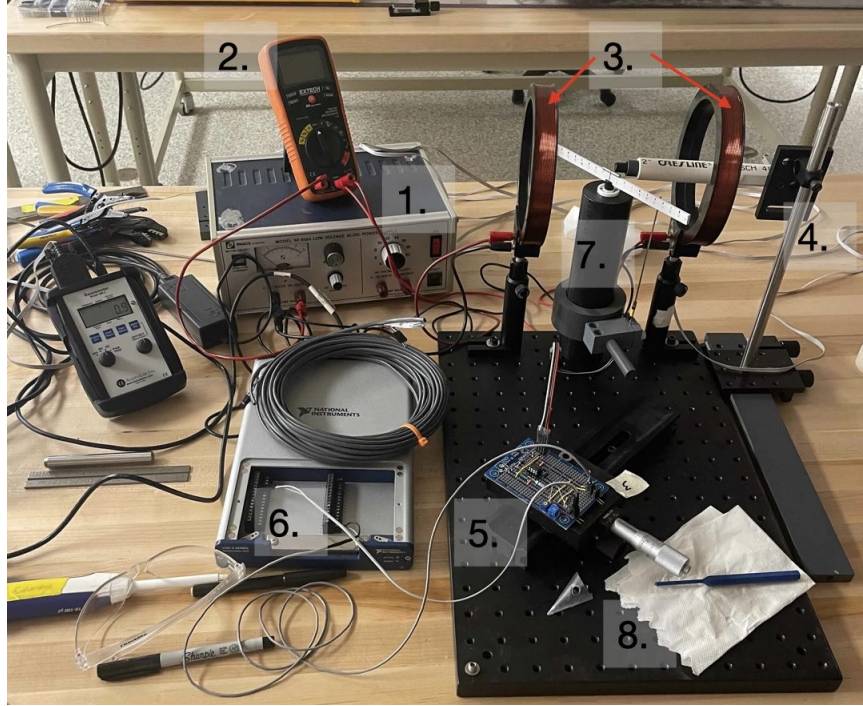


Figure 3.13: Experimental setup for torque. Includes magnetic field generation and measurement devices, 1 through 4. 5 is the circuit explained in section 3.3.2, 6. is a DAQ system connected to a laptop and 7. is the main device where the dowel subject sits.

In Clinic Measurement Protocol

Torque measurements were conducted ensuring consistency and reproducibility. Prior to setup, both researchers confirmed that all personal belongings and ferro-magnetic items had been removed from their pockets. The VI was opened with the necessary parameters configured, and the data acquisition card (DAC) was powered on. A 10-meter jumper wire was connected to the DAC and integrated into the circuit, with the computer system interfaced to the DAC for real-time data collection.

The experimental apparatus was then fitted with the test object—either the magnetically susceptible calibration material or the epicardial lead—mounted on the rotating ruler. Then it was positioned on the MRI room bench at a distance of 1 meter from the bore, measured from the base of the apparatus to the bore face. Finally, the

ruler arm was aligned to rest against the strain gauge, oriented as close as possible to 45° relative to the bore axis.

In the control room, torque measurements were acquired according to a predefined flowchart procedure, with results recorded on the measurement worksheet. A section of the worksheet was considered complete only when all data fields were marked with an “X.” Section 1 of the worksheet corresponded to measurements performed with the magnetically susceptible calibration object. Following these measurements, the apparatus was retrieved from the MRI room, and the calibration object was replaced with the epicardial lead affixed to the ruler. Section 2 of the worksheet was then completed by repeating the flowchart process with the lead in place.

Data collection in each section was performed at incremental positions along the MRI bore axis. For both the calibration object (Section 1) and the epicardial lead (Section 2), torque measurements were acquired at eleven translational positions: 100, 90, 80, 70, 60, 50, 40, 30, 20, 10, and 0 cm from the bore face. This systematic sampling allowed for direct comparison of torque response between the calibration object and the epicardial lead across the same magnetic field gradient.

3.5 Experimental Setup Summary

Chapter 4

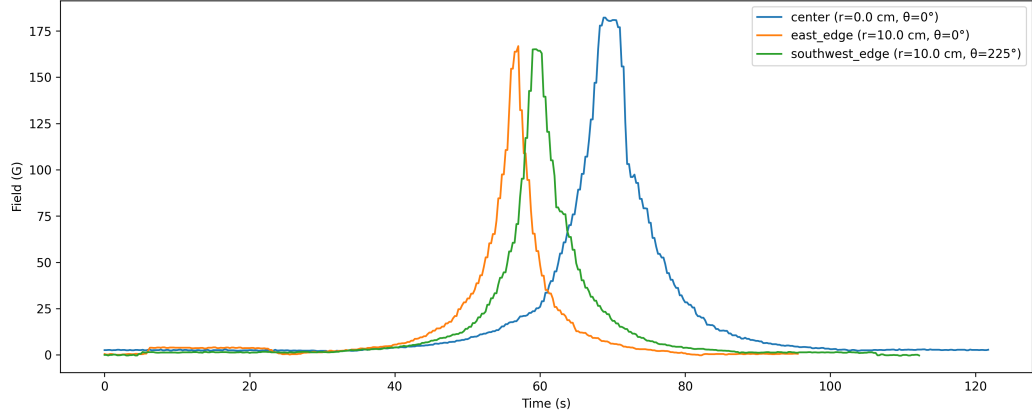
Results

overall description of the experiments and presentation of the data for the following:

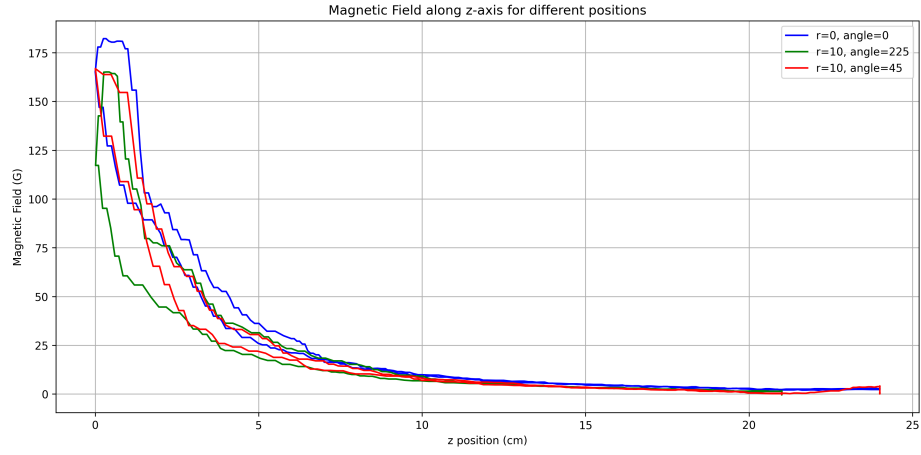
4.1 Magnetic Field Mapping

The magnetic field B generated in section 3.1, yielded to the following Figures. Figure 4.1 shows two plots from the field mapped, Fig. 4.1(a) shows the magnetic field flux B in gauss as a function of time going in and out to the solenoid front face, while Fig. 4.1(b) shows the estimated respective positions of the probe along the z -axis.

The three colors represent the position of the probe on each trial, this can be seen clearly on Figure 4.2 where the map was plotted as a representative line for their respective side. It is important to notice the similarities between runs on the southwest side and the east side of the front as this expected homogeneity allow us to extrapolate the data around the solenoid similar to Fig 2.10 confirming the theoretical background and speeding our clinical magnetic field acquisition protocol.



(a) B field as a function of time.



(b) B field as a function of z -position.

Figure 4.1: Magnetic field measurements plotted against time and axial displacement. Used to estimate stability and spatial variation of the mapping setup.

Figure 4.3 shows both the calculated gradient from equation 2.15, and the multiplication by field strength B . The later being useful on experiment 3.2 for finding magnetic force, as per equation 2.11, and for its fit (see Eq. 2.14)

3D Magnetic Field Magnitude from Measurements with Solenoid Circle

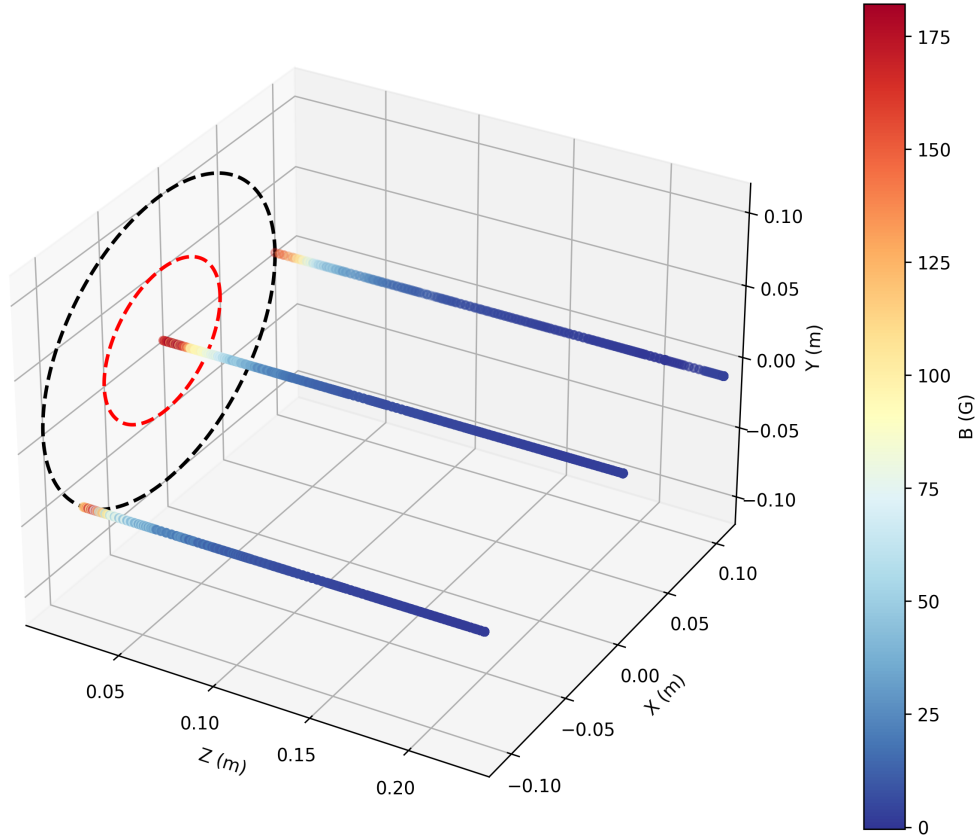


Figure 4.2: Three-dimensional reconstruction of the measured magnetic field, illustrating spatial variation across the mapping region.

4.2 Translational

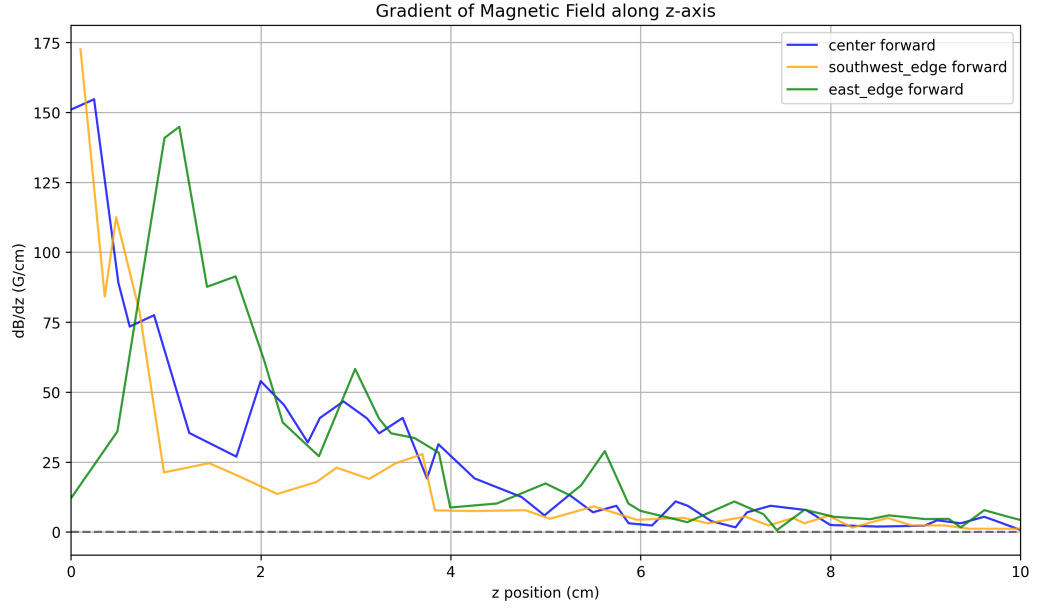
4.3 Torque

4.3.1 Sensor Calibration

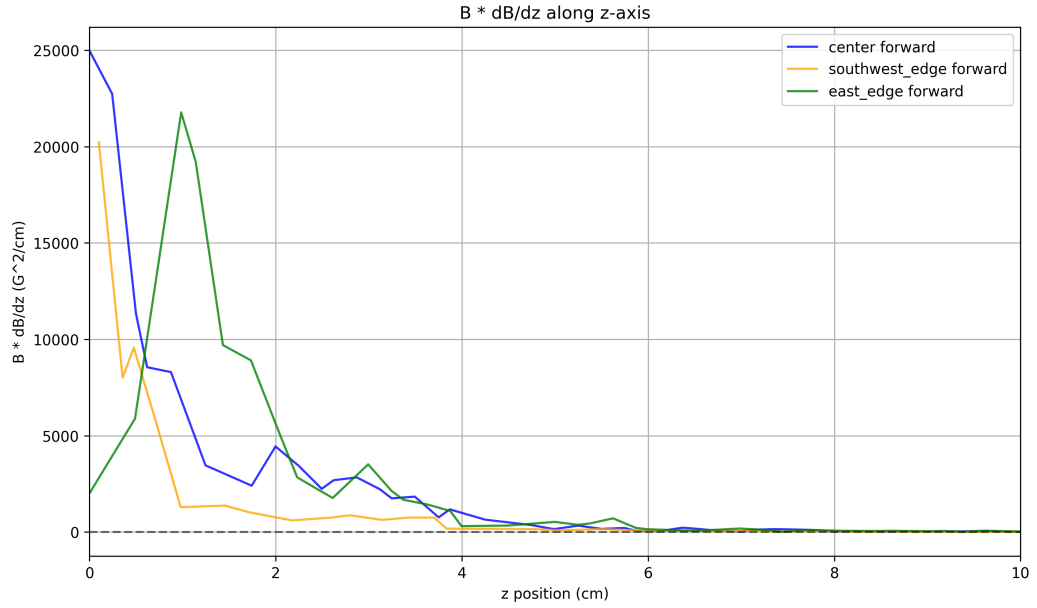
During calibration, aluminum based sensor was proven to be the best.

From Fig 4.5 the smallest measurable force the instrument can be determined. Using the slope of the most reliable fit of Aluminum rod $m = -3.4132e - 05(\leq 0.29\%), \chi_{red} = 1.05$.

The slope relates voltage changes to mass changes:



(a) ∇B along the measurement path.



(b) $B\nabla B$ product along the path.

Figure 4.3: Computed magnetic field gradients and the $B\nabla B$ product used for translational force characterization.

$$\Delta V = \text{slope} \cdot \Delta m$$

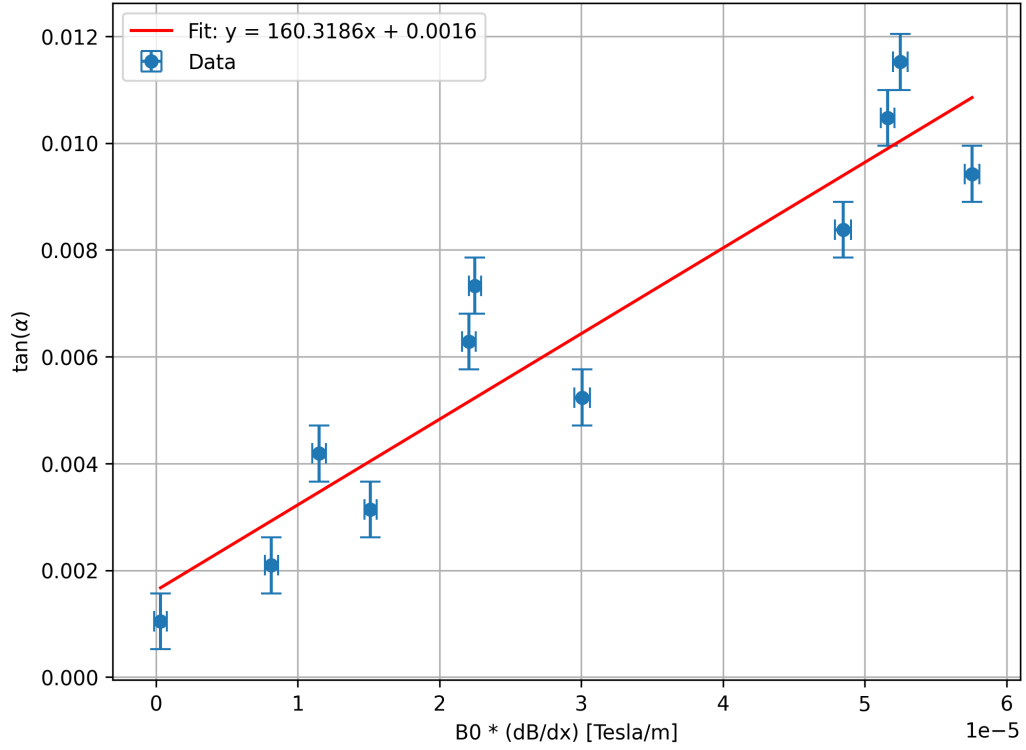
so the smallest measurable mass is

$$m_{min} = \frac{\Delta V}{|\text{slope}|}$$

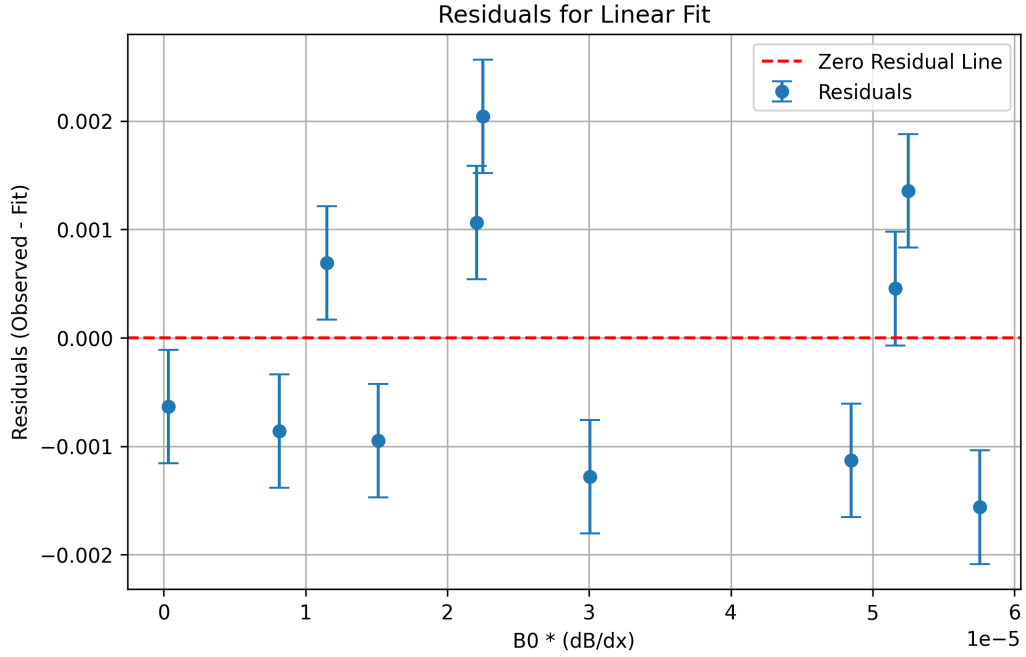
In this case, the standard error in voltage is $\Delta V = 1.4123e - 06$ which results in $m_{min} \sim 0.0414$ mg. This corresponds to a force of roughly $0.41\mu\text{Newton}$, which is significant because forces expected on an epicardial lead under a MRI magnetic field are extremely small.

Along with these measurements we also tested sensitivity along the length of sensor showing that the further out from the gauge bridge the more sensitive the sensor was. This is demonstrated on figure 4.7.

Figure 4.7 shows a small voltage for all distances from 50mm to 100mm and a range of voltages once we are high up (far from the bridge) with higher sensitivity near the center at 50mm.



(a) Curve fit of measured pendulum deflection vs. applied magnetic force.



(b) Residuals from linear fit of $\tan(\alpha)$ vs. $B_0 \cdot \frac{dB_0}{dz}$. Error bars indicate propagated uncertainties.

Figure 4.4: : (a) residuals from the linearized fit, and (b) fitted curve against the measured data points.

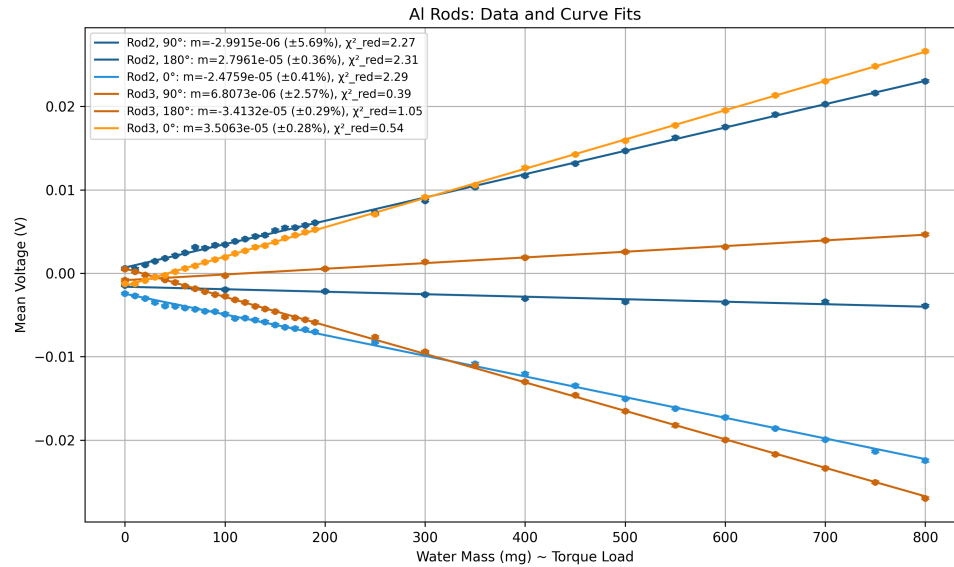


Figure 4.5: Calibration curves for two aluminum (AL) rods in three different orientations. Linear fits show consistent response across orientations, confirming the reliability of the lab-built sensor (see Methods Section for details).

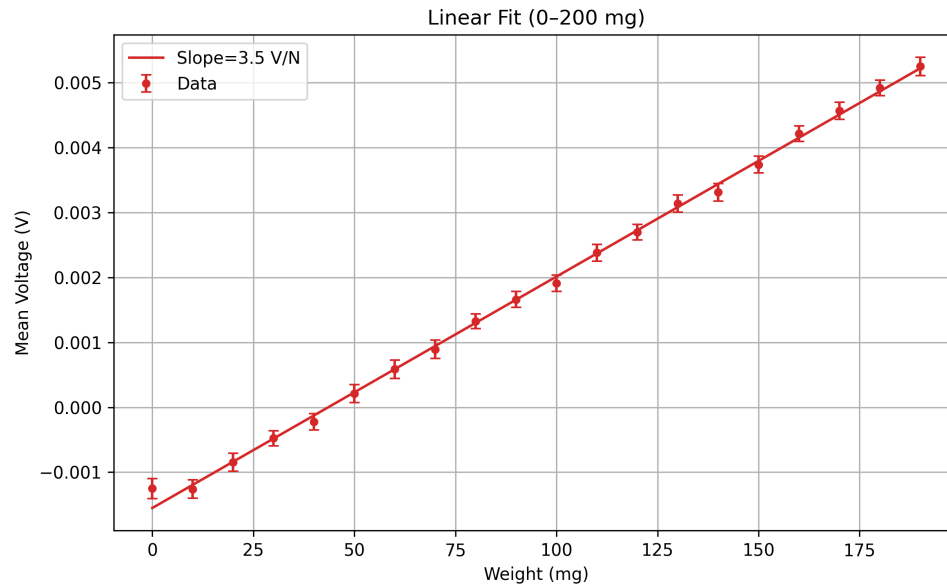


Figure 4.6:

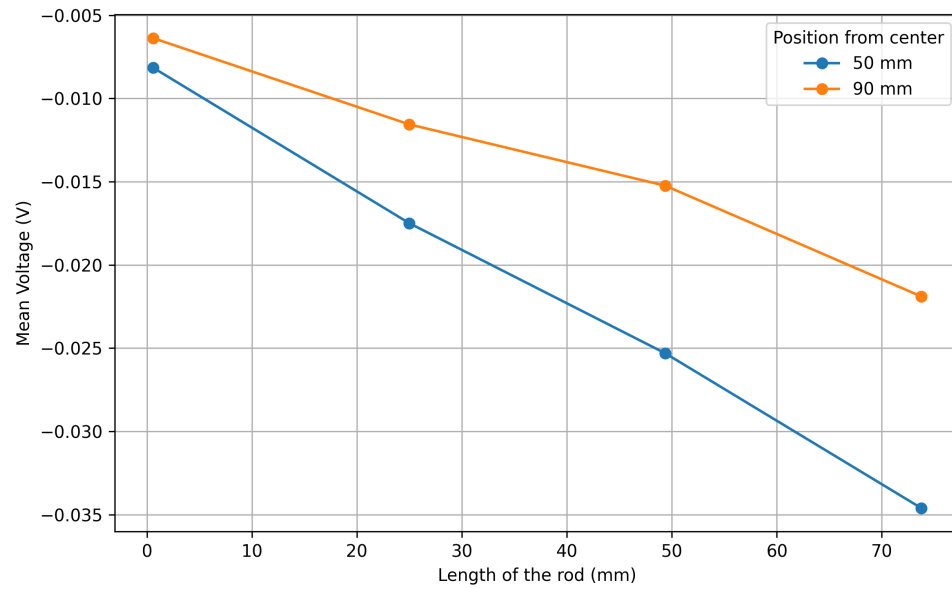
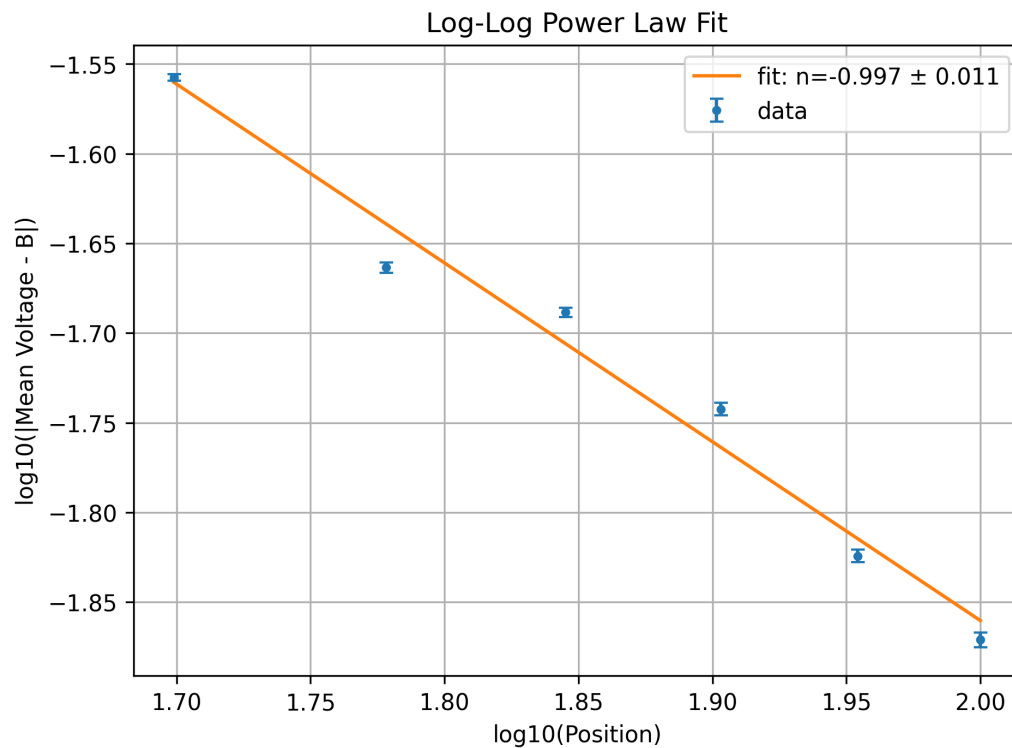
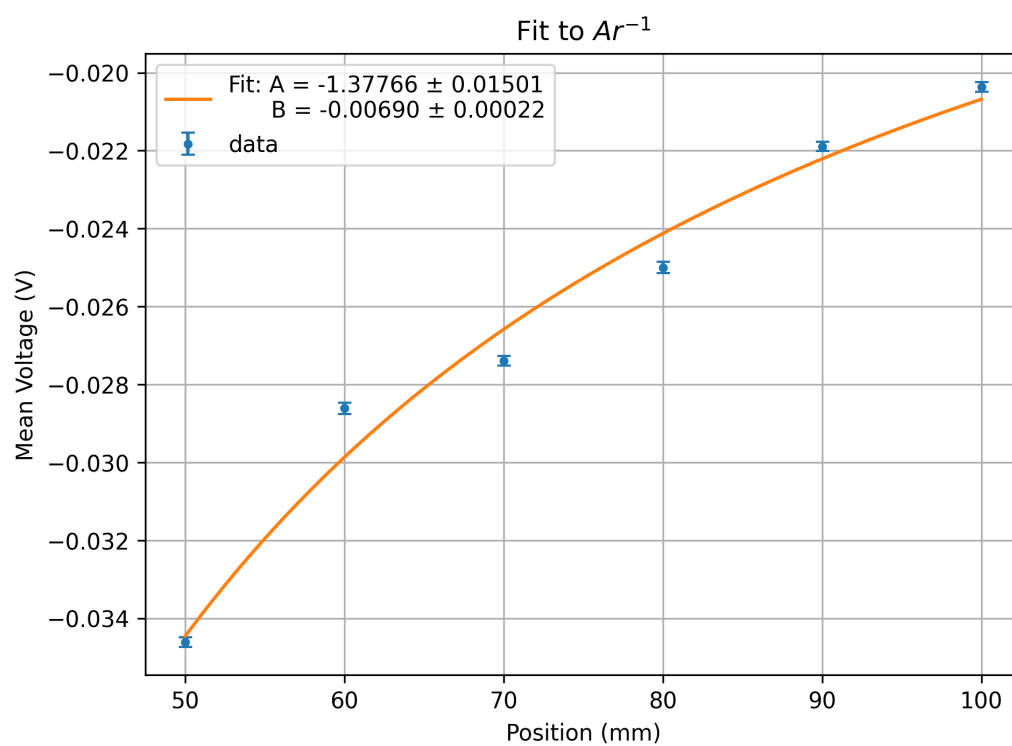


Figure 4.7: Mean voltage response of the sensor as a function of height for different distances from the center (50–100 mm). A small and relatively flat voltage is observed across all radii, while at larger heights (farther from the bridge) the measurements diverge, showing higher sensitivity near the center (50 mm). See Methods Section for details on torque transfer and distance selection.



(a)



(b)

Figure 4.8: Left: Ar^{-1} fit, force rises as the tip approaches the dipole center. Right: log-log verification.

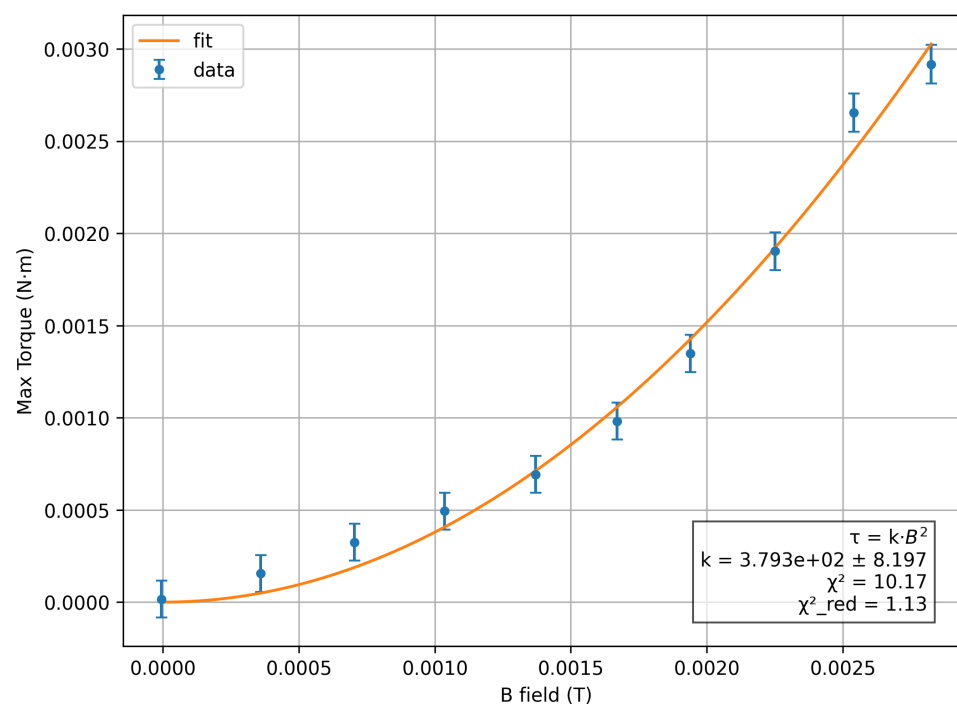


Figure 4.9:

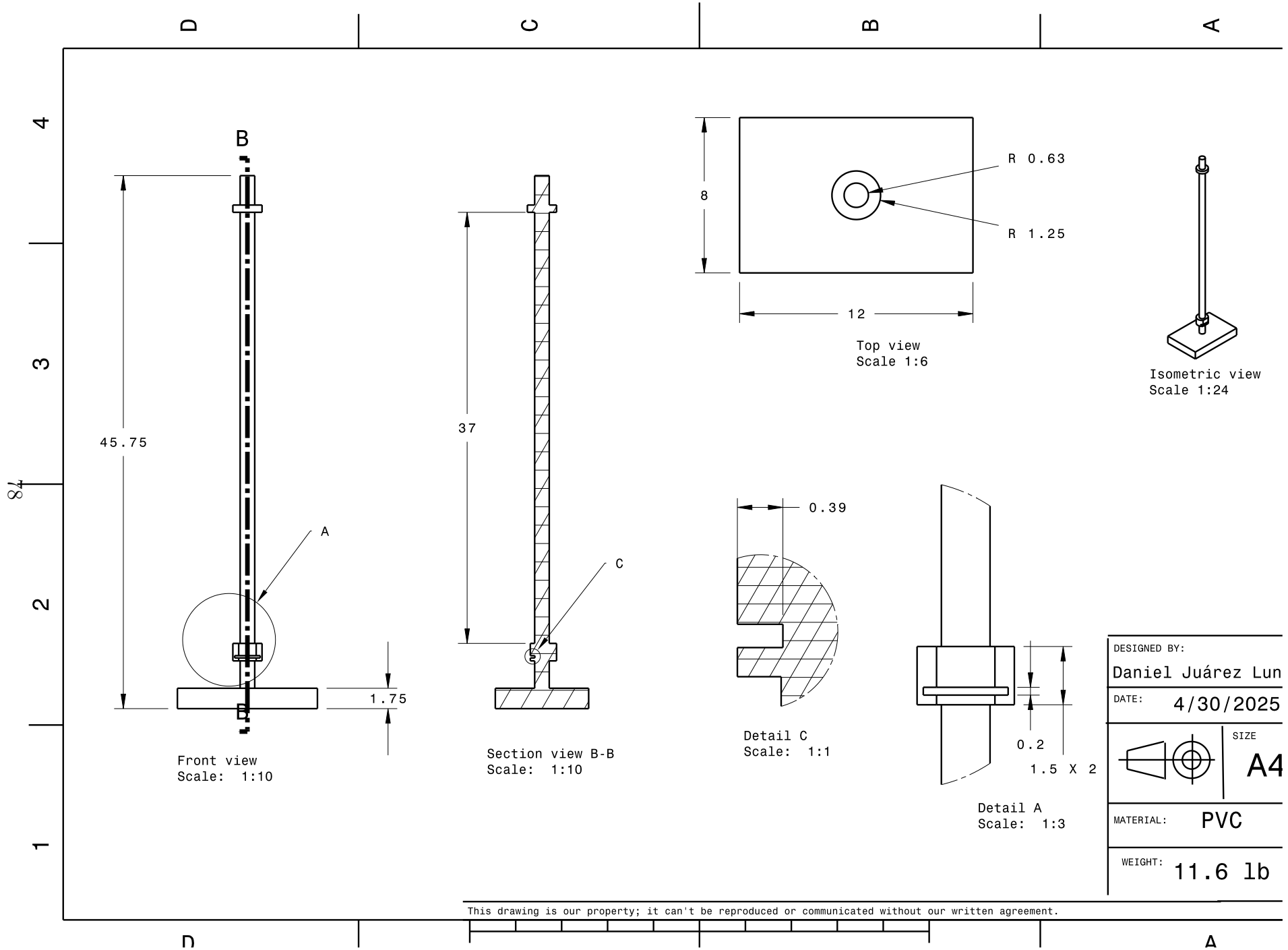
Chapter 5

Discussion and Conclusion

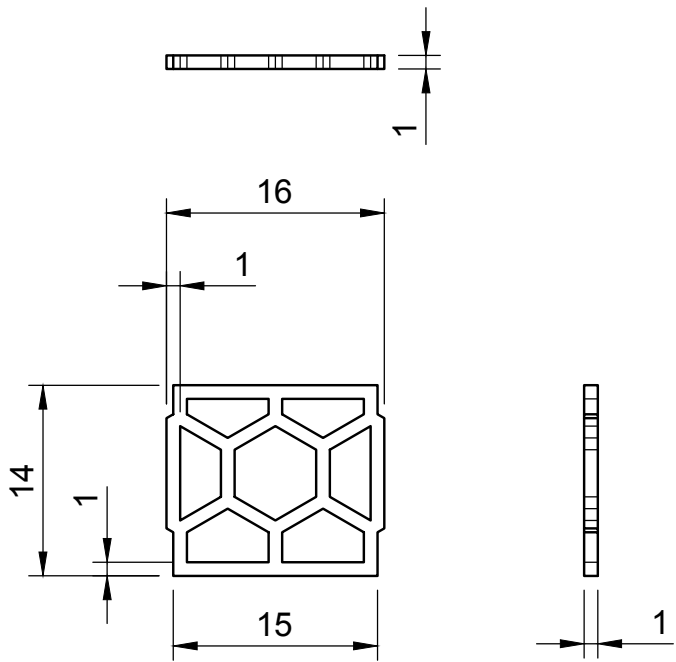
Appendix A

Engineering Drawings

Appendix A1: Pole drawing



Properties
Material: PLA
Weight:0.124 g



Dept.	Technical reference	Created by Daniel Juarez	8/5/25	Approved by
		Document type	Document status	
		Title Lead Holder Bee Comb Mesh MRI Translational Force	DWG No.	
		Rev.	Date of issue	Sheet 1/1

Appendix B

Sample calculations

B.1 Sample Calculation for Scaling at 30 G

Given that the MRI main field in the reference case is $B_0 = 3$ T, we consider a laboratory solenoid producing $B_0 = 30$ G = 3×10^{-3} T with a 5.691×10^{-3} T/m field gradient in z direction. The magnetic force acting on a weakly magnetic material is given by

$$\rho\chi = \frac{\mu_0 g \tan(\alpha)}{B_0 \frac{dB_0}{dz}}, \quad (\text{B.1})$$

where $\mu_0 = 1.256637 \times 10^{-6}$ N/A², $g = 9.8067$ m/s², ρ is the density of the laboratory material, and α is the equilibrium deflection angle of the pendulum, approximated by $\tan \alpha \approx 10/1000 = 0.01$ for a 1 cm displacement over a 1 m string.

For a scaled solenoid field of, assuming $B_0(dB_0/dz)$ scales quadratically with the magnetic field relative to the original MRI field of 3 T, we have

$$(B_0 \frac{dB_0}{dz})_{lab} = (B_0 \frac{dB_0}{dz})_{MRI} \left(\frac{B_{0,lab}}{B_{0,MRI}} \right)^2. \quad (\text{B.2})$$

Substituting the numerical values , the scaled gradient term for 30 G becomes

$$B_0 \frac{dB_0}{dz} = 1.71 \times 10^{-5} \text{ T}^2/\text{m}. \quad (\text{B.3})$$

Thus,

$$\rho\chi = \frac{(1.256637 \times 10^{-6})(9.8067)(0.01)}{1.71 \times 10^{-5}} = 7.22 \times 10^{-3}. \quad (\text{B.4})$$

Assuming $\rho = 7000 \text{ kg/m}^3$, the effective magnetic susceptibility for a ferromagnetic mimic becomes

$$\chi = \rho\chi \times \rho = (7.22 \times 10^{-3})(7000) = 5.05 \times 10^1 \approx 50. \quad (\text{B.5})$$

B.2 Smallest measurable force of translational fixture

Following section 2.2.5 For a specimen of $m = 0.1 \text{ g} = 1 \times 10^{-4} \text{ kg}$, the smallest measurable magnetic force is given by equation 2.21

$$F_{m,\min} = mg \frac{z}{L}, \quad (\text{Eq. 2.21})$$

where $g = 9.81 \text{ m/s}^2$, $z = 1 \times 10^{-3} \text{ m}$, and $L = 1 \text{ m}$. Substituting these values yields

$$\begin{aligned} F_{m,\min} &= (1 \times 10^{-4})(9.81) \left(\frac{1 \times 10^{-3}}{1} \right) \\ &= 9.81 \times 10^{-7} \text{ N} \approx 1.0 \times 10^{-6} \text{ N}. \end{aligned} \quad (\text{B.6})$$

Thus, for a 0.1 g mass, the smallest detectable magnetic force is approximately $1 \mu\text{N}$. The corresponding force ratio remains

$$\frac{F_m}{F_g} = \frac{z}{L} = 10^{-3}, \quad (\text{B.7})$$

indicating that the setup can detect magnetic forces as small as one-thousandth of the object's weight.

B.3 Smallest measurable force of translational fixture

B.4 Kirchhoff analysis of Circuit

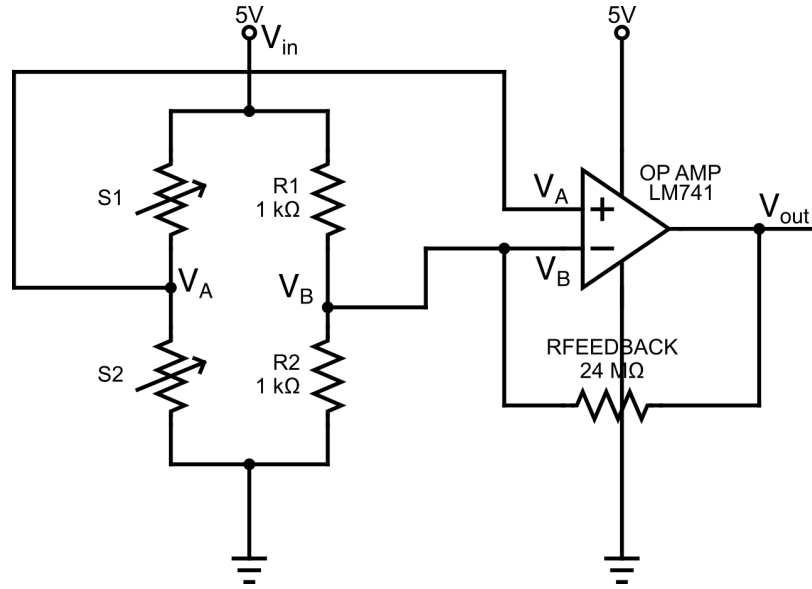


Figure B.1: Schematic of the Wheatstone bridge with strain gauges $S1$ and $S2$, fixed resistors $R = R1 = R2$, and operational amplifier with feedback resistor R_f . This configuration corresponds to the transfer function in Equation (3.3).

Using Kirchhoff's Current Law (KCL), in Figure B.1 for node V_A which corresponds to the positive leg of the op amp, or non inverting voltage, we have

$$\begin{aligned}
\frac{V_A - V_{in}}{S1} + \frac{V_A}{S2} &= 0 \\
\frac{S2(V_A - V_{in}) + S1V_A}{S1S2} &= 0 \\
V_A(S2 + S1) - S2V_{in} &= 0 \\
V_A &= \frac{S2V_{in}}{S2 + S1}.
\end{aligned}$$

Similarly, for the node V_B corresponding to the inverting input of the op-amp

$$\begin{aligned}
\frac{V_B - V_{in}}{R} + \frac{V_B}{R} + \frac{V_B - V_{out}}{R_f} &= 0 \\
\frac{R_f(2V_B - V_{in}) + R(V_B - V_{out})}{R\cancel{R_f}} &= 0 \\
V_B(2R_f + R) - R_fV_{in} - RV_{out} &= 0 \\
V_B &= \frac{R_fV_{in} + RV_{out}}{2R_f + R}.
\end{aligned}$$

At this point, we make a standard ideal op-amp assumption, the op-amp has infinite open-loop gain, meaning any finite output voltage corresponds to a vanishingly small input voltage difference. That is, for the difference amplifier

$$V_{out} = A_{ol}(V_A - V_B), \quad A_{ol} \rightarrow \infty, \quad (\text{B.8})$$

which implies $V_A \approx V_B$. Substituting $V_A = V_B$ into the previous equations allows us to solve for the closed-loop transfer function

$$\begin{aligned}
\frac{S2V_{in}}{S2 + S1} &= \frac{R_f V_{in} + R V_{out}}{2R_f + R} \\
\frac{V_{out}}{V_{in}} &= \frac{R_f(S2 - S1) + S2R}{R(S2 + S1)} \\
\frac{V_{out}}{V_{in}} &= \frac{R_f}{R} \frac{S2 - S1}{S2 + S1} + \frac{S2}{S2 + S1}
\end{aligned} \tag{3.3}$$

In summary, the output voltage is composed of two contributions: a differential term amplified by the ratio R_f/R , which captures the imbalance between the strain gauges $S1$ and $S2$, and a baseline voltage division term $S2/(S2 + S1)$ that represents the inherent bridge offset. This expression clearly shows how the Wheatstone bridge imbalance is amplified by the op-amp to produce the measurable output.

B.5 Explaining the behavior of Coarse-Fine Tuning with Potentiometers in Circuit

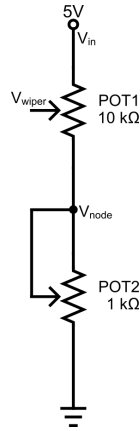


Figure B.2: Schematic of the coarse-fine potentiometer configuration used for the voltage-divider derivations in Figure 3.9.

B.5.1 1. Derivation of the Node Voltage

Starting with KVL around the loop

$$V_{\text{in}} - IR_1 - IR_2 = 0, \quad (\text{B.9})$$

Factor out the current

$$V_{\text{in}} = I(R_1 + R_2), \quad (\text{B.10})$$

Solve for the series current

$$I = \frac{V_{\text{in}}}{R_1 + R_2}, \quad (\text{B.11})$$

The node voltage is the drop across the lower resistor R_2

$$V_{\text{node}} = I R_2 = \left(\frac{V_{\text{in}}}{R_1 + R_2} \right) R_2. \quad (\text{B.12})$$

Final voltage-divider form

$$\boxed{V_{\text{node}} = V_{\text{in}} \frac{R_2}{R_1 + R_2}}. \quad (\text{3.4})$$

2. Numerical Evaluation for Fine Pot Positions

For these evaluations

$$V_{\text{in}} = 5.0 \text{ V}, \quad R_1 = 10 \text{ k}\Omega, \quad R_2 \in [0, 1000] \text{ }\Omega. \quad (\text{B.13})$$

2.1 Halfway Fine Adjustment $R_2 = 500 \text{ }\Omega$

$$V_{\text{node}} = 5.0 \frac{500}{10000 + 500} = 5.0 \left(\frac{500}{10500} \right) = 5.0(0.047619) = 0.2381 \text{ V}. \quad (\text{B.14})$$

2.2 Tabulated Values for Fine Pot Settings

$$V_{\text{node}} = 5 \frac{R_2}{10000 + R_2}. \quad (\text{B.15})$$

Fine Pot Setting	R_2 (Ohms)	V_{node} (V)
Max	1000	0.4545
Mid	500	0.2381
Min	0	0

3. Coarse Adjustment with Fine Fixed at Midpoint

Now fix the fine pot at

$$R_2 = 500 \, \Omega. \quad (\text{B.16})$$

The 10 k coarse pot wiper divides R_1 into two portions. Let the coarse wiper position be $\alpha \in [0, 1]$, where

$$\alpha = 0 : \text{wiper at top}, \quad \alpha = 1 : \text{wiper at bottom}. \quad (\text{B.17})$$

The wiper voltage is a weighted average between the supply (5 V) and the node voltage computed with the fixed fine setting

$$V_{\text{node, fine}=500} = 0.2381 \, \text{V}. \quad (\text{B.18})$$

Thus the coarse-wiper voltage is

$$V_{\text{wiper}} = (1 - \alpha)(5.0) + \alpha(0.2381). \quad (\text{B.19})$$

3.1 Coarse-Wiper Table (Fine Fixed at Half)

Coarse Setting	α	V_{wiper} (V)
Top (Min Coarse)	0	5.000
Midpoint	0.5	2.619
Bottom (Max Coarse)	1	0.238

4. Interpretation

At $\alpha = 0$ (coarse at minimum), the wiper sits at the full input voltage

$$V_{\text{wiper}} = 5 \text{ V}, \quad (\text{B.20})$$

which is independent of the fine adjustment. At $\alpha = 1$ (coarse at maximum), the wiper voltage collapses into the range controlled solely by the fine pot

$$V_{\text{wiper}} = V_{\text{node, fine}}. \quad (\text{B.21})$$

Thus the coarse pot sets the overall span, and the fine pot determines the precise lower-region tuning.

Appendix C

Materials and Components

Reference Sheet

Appendix D

Calibration Logs

Additional experimental relationships and fitting diagnostics are presented to support the main analysis.

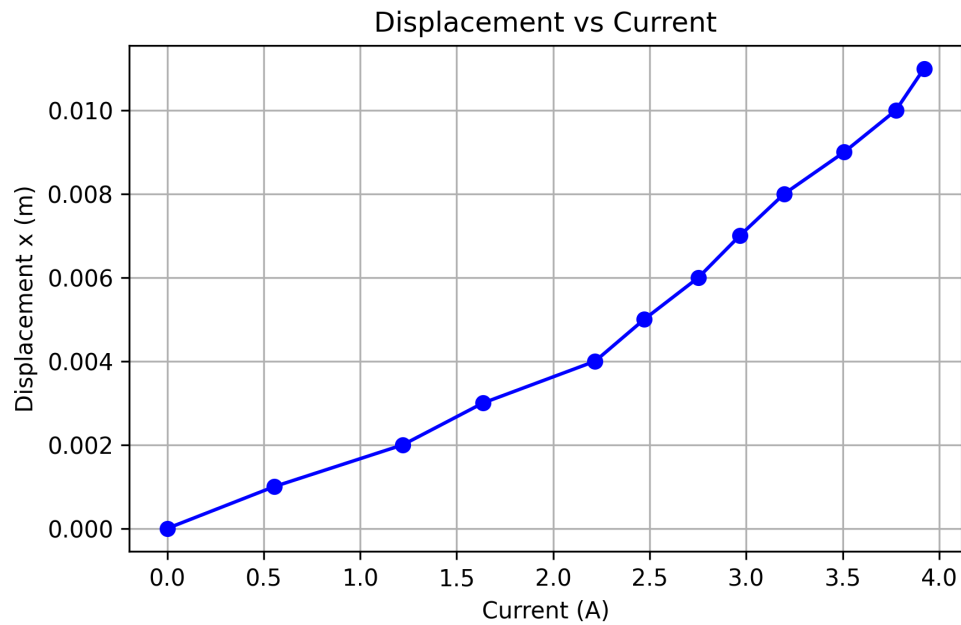


Figure B.1: Displacement as a function of current through the solenoid.

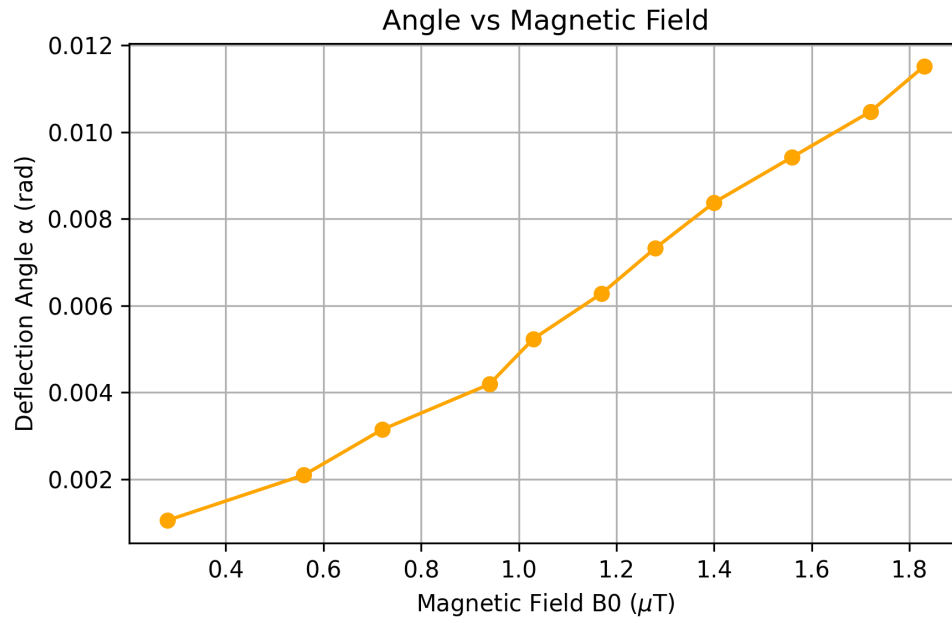


Figure B.2: Measured angle α versus magnetic field intensity B_0 .

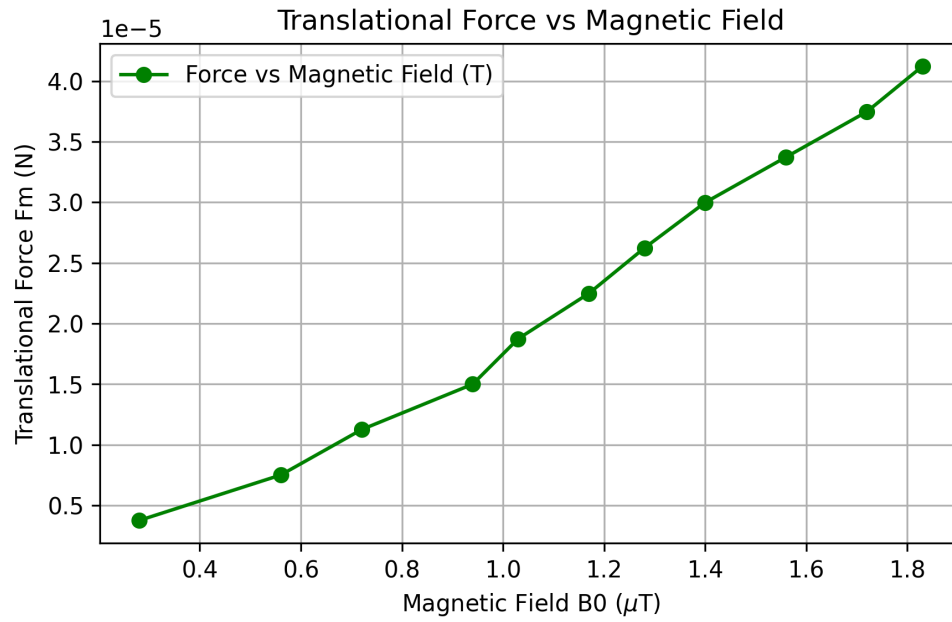


Figure B.3: Computed translational force F_m as a function of B_0 .

Appendix E

Codes and data samples

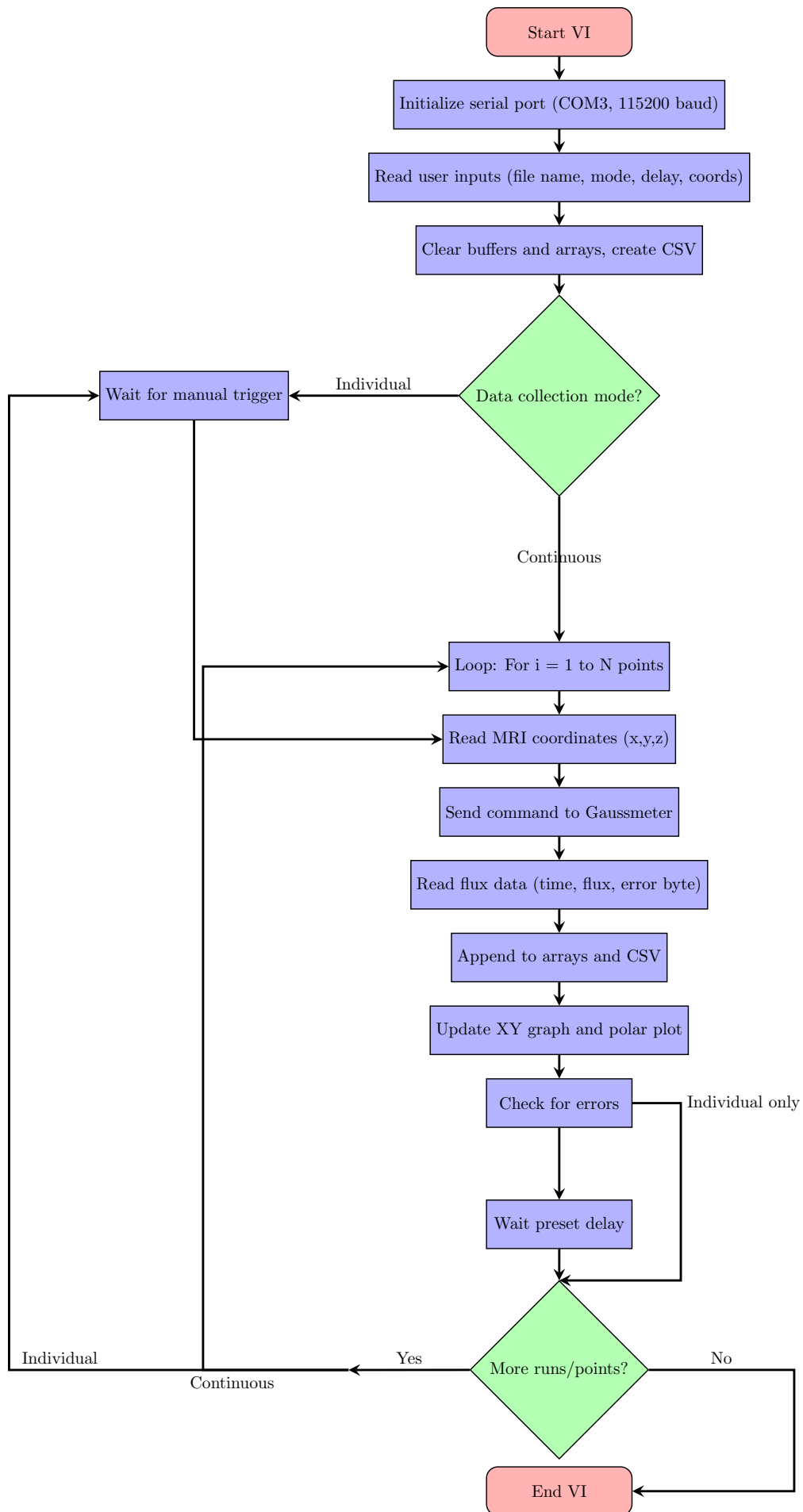
E.1 Gaussmeter LabView Protocol

Figure E.1. Upon startup, the VI initializes serial communication with the Gaussmeter, loads user-defined parameters (such as file name, mode of acquisition, delay time, and probe coordinates), and clears buffers in preparation for logging. The user may select either a continuous or an individual sampling mode. In continuous mode, the VI iterates through a defined number of data points with a fixed delay between samples; in individual mode, each data capture is triggered manually, allowing for precise positioning or timing. For both modes, each cycle involves reading the spatial coordinates, sending a measurement command, recording the returned magnetic flux value, and updating a graphical display as well as the output file.

While the flowchart reflects the core logic, it does not fully convey the customization added to the VI interface. For example, the front panel includes input fields for patient table (couch) parameters such as couch speed and displacement points. As mentioned above, phantom is discretized with radial and angular indices: angular positions span 12 sectors clockwise from the 12 o'clock position (0 to 11 or A to L), and radial depths range from the center (0) to the outer edge (6). The probe's spatial

location is represented in polar coordinates, and the system plots magnetic flux as a function of position rather than time, allowing for intuitive visualization of the MRI fringe field across the phantom surface. The VI is operated in the following way.

Flowchart of the Gaussmeter VI data acquisition algorithm



E.2 Sample Data for Translational Force Experiment

Table E.1 presents a representative subset of the displacement–current dataset used for field–gradient and angle calculations for experiment en section 3.2.2 Uncertainties: position ± 0.05 mm, current ± 0.001 A.

N	x [mm]	I [A]	$B_{-0.5}$	B_0	$B_{+0.5}$	dB/dx	θ	$B_0 dB/dx$
1	0	0.000	–	–	–	–	–	–
2	1	0.554	2.86	2.81	2.84	0.1111	0.001047	0.3123
3	2	1.221	5.56	5.66	5.70	1.4444	0.002094	8.1691
4	3	1.636	7.03	7.22	7.24	2.1111	0.003141	15.2469
5	4	2.217	9.32	9.44	9.44	1.2222	0.004188	11.5432
6	5	2.471	10.27	10.40	10.56	2.8889	0.005235	30.0444
7	6	2.752	11.61	11.66	11.80	1.8889	0.006282	22.0160
8	7	2.966	12.58	12.63	12.76	1.7778	0.007329	22.4593
9	8	3.196	13.92	14.02	14.27	3.4444	0.008377	48.2988
10	9	3.507	15.52	15.69	15.89	3.6667	0.009424	57.5259
11	10	3.775	17.01	17.28	17.31	3.0000	0.010471	51.8333
12	11	3.921	18.03	18.17	18.32	2.8889	0.011518	52.4815

Table E.1: Representative dataset of displacement, current, magnetic field samples at ± 0.5 mm and equilibrium, computed gradients, and corresponding angular measurements.

Appendix F

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Glossary

AIMD active implantable medical devices. 6, 8

CDR Creighton Digital Repository. 95

CIED cardiac implantable electronic device. 1, 2, 12

CRT cardiac resynchronization therapy. 12

DAC data acquisition card. 64

DAQ Data Acquisition. xii, 60, 64

ECG electrocardiogram. 1

FEG Frequency Encoding Gradient. 21, 22

HRS Heart Rhythm Society. 1

ICD implantable cardioverter-defibrillators. 12

KCL Kirchhoff's Current Law. 82

MRI Magnetic Resonance Imaging. x, 1, 2, 3, 4, 7, 12, 17, 18, 20, 21, 22, 23, 24, 25,
30, 42, 43, 45, 46, 50, 70

PD Proton Density. 17

PEG Phase Encoding Gradient. 21, 22

PLA Polylactic Acid. 48

PVC Polyvinyl Chloride. 9, 45, 47, 54

RF radiofrequency. 1, 2, 3, 7, 17

SSG Slice Select Gradient. 20, 22

VI LabVIEW Virtual Instrument. 46, 60, 64, 91, 92

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